

University of Western Australia

Master of Physics (Computational Physics)

Quantum Circuits for Solving Differential Equations via Physics-Informed Effective Hamiltonians and GQSP Quantum Imaginary Time Evolution

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9 October 2026

Declaration

This is to certify that:

1. This thesis comprises of my original work.
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Acknowledgments

Thank you to my supervisor, Jingbo Wang, for her guidance and support throughout my research, particularly for her continual encouragement and belief in me.

I would also like to thank my family and friends for their support and encouragement throughout my studies.

Summary of Student Achievement

Abstract

Brief summary.....

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Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Discretisation

Symbol	Meaning	Introduced
n	Qubits per Chebyshev axis (the resolution).	§ 3.1
$N = 2^n$	Truncation dimension of one Chebyshev axis.	§ 3.1
k	Order of an ordinary differential equation.	§ 3.2
p	Degree of a polynomial nonlinearity.	§ 7.3
n_f	Qubits in the field (block) register of the descriptor form.	§ 5.3
$\dim w$	Padded descriptor state dimension, $2^{\lceil \log_2(k+1) \rceil} N$.	Appendix G.2

The Chebyshev model

Symbol	Meaning	Introduced
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ.	§ 3.1
$\lvert\tau(x)\rangle$	The raw Chebyshev feature state.	§ 3.1
ψ, η, η_e	Normalised coefficient state; its scale; $\eta_e = \lVert b \rVert^2$.	§ 3.1
$\hat{f}^{(j)}$	Coefficients of $f^{(j)}$: one block of the descriptor state.	§ 5.3
w	The descriptor state $(\hat{f}^{(0)}; \ldots; \hat{f}^{(k)})$.	§ 5.3

Operators and constraints

Symbol	Meaning	Introduced
$\mathbb{G}$	Differentiation matrix, $f \mapsto f'$, in coefficient space: dense, strictly upper triangular, $\|\mathbb{G}\|_2 = \Theta(N^2)$ (Wu et al. print it as $\mathbb{G}_n^\top$).	§ 3.2
$\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$	Banded factors of the differentiation matrix, $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$.	§ 5.2
$\mathbb{M}_{x^p}$, $\mathbb{M}_a$	Multiplication by x^p (the exact lift, $2N \times N$) and by a smooth $a(x)$.	§ 3.2
$\mathbb{n}_1$, $\mathbb{n}_x$	Product-to-sum folds.	§ 3.4
$\mathbb{B}(x)$, $\mathbb{D}^{(0)}(x_s)$	Rank-one zero (invariant) and collapse (regular) constraint rows.	§ 3.2
x_z, x_m, x_s	Constraint points: a zero, an extremum, a source datum.	§ 3.2
R_0, U_{odd}, Λ	Factors of the structured exact encoding of $\mathbb{G}$.	§ 4.5
$\mathbb{S}_a(x^*)$, $\hat{g}$	Datum-slice row on axis a at x^*; the coefficients of the datum function.	§ 4.3
$\mathbb{U}$	The Chebyshev–Gauss transform used by the nodal fold.	§ 7.4
$\|\mathbb{G}\|_S$	The paper's norm $\max\big(\|\mathbb{G}\mathbb{G}^\top\|_1, \|\mathbb{G}^\top\mathbb{G}\|_1\big)$.	§ 3.3

Residuals and Hamiltonians

Symbol	Meaning	Introduced
A_{std}, A_{desc}	Residual of the standard form; of the descriptor form.	§ 3.2
H_{std}, H_{desc}	Effective Hamiltonian of each form; bare H only in regime-independent statements.	§ 3.2
$R = [R_1; \ldots; R_L]$	Stacked residual, with $H = R^\dagger R$.	§ 8.3
$H_\otimes$	The doubled-space Hamiltonian.	§ 7.2

Block encoding, filtering and state preparation

Symbol	Meaning	Introduced
α, α_R, $\alpha_H = \alpha_R^2$	Subnormalisation: generic, of R, of H.	§ 2.4
V, W	Stacked-residual reflection; qubitised walk.	§ 8.3
$H' = 2H/\alpha_R^2 - I$	The normalised Hamiltonian that V block-encodes.	§ 8.3
Δ	Spectral gap $\lambda_1 - \lambda_0$.	§ 8.4
β	Imaginary time.	§ 8.4
d, q_k	GQSP degree; filter coefficients.	§ 8.4
ε	Target infidelity (10^{-6}).	§ 8.4
φ_0, γ	Initial state; its overlap $\lvert \langle \psi_{\text{ref}} \rvert \varphi_0 \rangle \rvert$ with the reference.	§ 8.4
p_G	GQSP success probability.	§ 8.4
ζ	Block-rescaling constant.	§ 5.6
$\epsilon_{\mathbb{G}}$, μ	Entry error and per-bit coefficient of the published encoding of $\mathbb{G}$.	Appendix D.2

Nonlinearity

Symbol	Meaning	Introduced
K	Carleman truncation order.	§ 7.3
M	Dimension of the base state u.	§ 7.3
ρ	Nonlinearity-to-dissipation ratio; the Carleman precondition is $\rho < 1$.	§ 7.3
F_1, F_2, $\mathbb{A}_K$	Linear and quadratic terms; the lifted Carleman generator.	§ 7.3

Systems and complex coefficients

Symbol	Meaning	Introduced
C	Number of components (fields) of a system.	§ 6.1
z	A system's state: one block per component.	§ 6.1
$\mathbb{A}$	A system's coupling matrix, $u_i' = \sum_j \mathbb{A}_{ij} u_j$.	§ 6.1
$\mathbb{F}$	Ratio row: fixes the direction of a vector datum and leaves its scale free.	§ 6.2
y, y_a, $y_\perp$	The vector datum; its anchor component; its off-anchor part.	§ 6.2
w_0	A system's stated initial data; it weights the initial state.	§ 6.2
$c = \operatorname{Re} c + i \operatorname{Im} c$	A complex coefficient, in the realification example.	§ 6.5

Acronyms

GQSP generalised quantum signal processing.

LCU linear combination of unitaries.

NLFT nonlinear Fourier transform.

ODE ordinary differential equation.

PDE partial differential equation.

PINN physics-informed neural network.

QITE quantum imaginary-time evolution.

QSVT quantum singular value transformation.

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 Physics-Informed Hamiltonians and Where Their Cost Lies

1.3 Contributions and Scope

1.4 Outline

2 Background and Literature Review

2.1 Quantum Computation Preliminaries

In classical computing, the fundamental unit of information is the bit, which can take one of two values: 0 or 1.

The fundamental unit of information in quantum computing is the qubit, which can exist in a superposition of states, allowing it to encode both 0 and 1 simultaneously. This property enables quantum computers to perform certain computations more efficiently than classical computers.

Definition 1 (State Space): The set of all quantum states that a quantum system can occupy is a complex vector space with inner product structure known as a Hilbert space (denoted $\mathcal{H}$). A finite dimensional Hilbert space is isomorphic to $\mathbb{C}^N$ ($\mathcal{H} \cong \mathbb{C}^N$) for some $N \in \mathbb{N}$, where N is the dimension of the Hilbert space. A quantum state is a unit vector in this Hilbert space, represented in dirac notation as $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$ (normalisation condition). A quantum state $|\psi\rangle$ can be expressed as a linear combination of basis states $\{|i\rangle\}_{i=0}^{N-1}$, where $|i\rangle$ is the ith basis state of the Hilbert space.

$$|\psi\rangle = \sum_{i=0}^{N-1} \psi_i |i\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \tag{2.1}$$

Its hermitian conjugate (or adjoint) is denoted as $\langle\psi| = |\psi\rangle^\dagger$. Note that for some $e^{i\theta}$ for $\theta \in [0, 2\pi)$, the states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically indistinguishable, and thus correspond to the same physical quantum state. This is known as a global phase factor.

Definition 2 (The Qubit): A quantum bit (qubit) is a two-state quantum system, characterised by $|\psi\rangle \in \mathbb{C}^2$.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \tag{2.2}$$

where $\alpha, \beta \in \mathbb{C}$, $|0\rangle$ and $|1\rangle$ are the computational basis states, and $|\alpha|^2, |\beta|^2$ are the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively. The normalisation condition requires that $|\alpha|^2 + |\beta|^2 = 1$.

Definition 3 (Composite Systems): The state space of a composite quantum system is the tensor product of the state spaces of its subsystems. Consider two quantum states $|\psi\rangle_A \in \mathcal{H}_A$

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and $|\psi\rangle_B \in \mathcal{H}_B$, the composite state $|\psi\rangle_{AB}$ is given by:

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \tag{2.3}$$

where $\otimes$ denotes the tensor product and is often hidden by convention. The dimension of the composite state space is the product of the dimensions of the individual state spaces. If a composite system cannot be written as a tensor product of its subsystems, it is said to be entangled. Entanglement is a unique feature of quantum mechanics that has no classical counterpart and plays a crucial role in quantum information processing.

We often consider the state of a composite system of n qubits, which is represented in a 2^n-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{2^n}$.

2.1.1 Quantum Gates and Circuits

Definition 4 (Quantum Algorithm): A quantum algorithm is a sequence of operations that manipulates qubits to perform a specific computation.

The goal of any quantum algorithm is to manipulate quantum states in a controlled manner to achieve a desired outcome. This is accomplished through the application of quantum gates, which are unitary operations that transform quantum states. Quantum gates can be combined to form quantum circuits, which represent the sequence of operations performed on qubits. Each quantum gate can be implemented using physical operations on hardware, and the design of efficient quantum algorithms is a central challenge in quantum computing research.

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Definition 5 (Unitary Evolution): The evolution of a closed quantum system between two states is described by unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, where n is the number of qubits in the system.

$$|\psi\rangle_{final} = U\,|\psi\rangle_{initial}\,, \quad U^\dagger U = UU^\dagger = I \tag{2.4}$$

where $U^\dagger$ is the hermitian conjugate of U and I is the identity operator. Unitary transformations preserve the inner product and thus the norm of quantum states, ensuring that states have a physical interpretation as probability amplitudes.

These unitary transformations are known as quantum gates, and each gate serves a specific purpose in manipulating quantum states.

The quantum circuit diagram offers a succinct and intuitive representation of quantum algorithms, analogous to Feynman diagrams in quantum field theory.

Time flows left to right, and each wire (horizontal line) represents a qubit or a quantum register. Gates are placed sequentially on the wires indicating operations applied to qubits. An example is the Hadamard gate, which creates superposition, and the CNOT gate, which entangles qubits. A full list of quantum gates and their corresponding circuit symbols can be found in .

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2.1.2 Complexity of Quantum Algorithms

Definition 6 (Gate Complexity): The total number of quantum gates used in a quantum circuit.

Definition 7 (Circuit Depth): The maximum number of gates applied sequentially on any qubit in a quantum circuit.

Definition 8 (Ancillary Qubits): Additional qubits introduced into a quantum circuit beyond the primary input qubits, typically initialised to a baseline state (such as $|0\rangle$) to assist with intermediate calculations, temporary storage, or entanglement generation.

Definition 9 (Big-$\mathcal{O}$ Notation): Big-$\mathcal{O}$ notation is a mathematical notation used to describe the asymptotic behaviour of functions.

$$\mathcal{O}(f(n)) = \{g(n) : \exists c > 0, n_0 \in \mathbb{N} \text{ such that } 0 \leq g(n) \leq cf(n) \text{ for all } n \geq n_0\} \tag{2.5}$$

In the context of quantum algorithms, it is used to express the scaling of gate complexity, circuit depth, and space requirements with respect to the number of qubits n.

For example, if a quantum algorithm has a gate complexity of $\mathcal{O}(n^2)$, it means that the number of gates required grows quadratically with the number of qubits.

A quantum circuit is said to be **efficient** if its circuit depth, gate complexity, and the number of ancillary qubits all scale polynomially with the number of qubits n.

2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era

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Table 1: Quantum approaches to differential equations: what each assumes, what it costs, and where it fails.

Approach	Assumptions	Query cost	Output	Failure mode
Linear systems (HHL family)				
Linear combination of Hamiltonian simulations				
Carleman linearisation				
Variational				
Physics-informed				
Hamiltonian [1]				
This work				

2.2 Quantum Algorithms for Differential Equations

2.2.1 Linear-Systems and Linear-Combination Routes

2.2.2 Nonlinear Equations: Carleman and Related Linearisations

2.2.3 Variational Routes

2.2.4 Ground-State Routes and the Physics-Informed Lineage

2.2.5 Critical Assessment

2.3 Spectral Methods and Their Conditioning

2.3.1 Chebyshev Series, Galerkin Truncation and Differentiation

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2.4 Block Encoding, Linear Combinations of Unitaries and Qubitisation

2.5 Quantum Signal Processing and Its Generalisation

2.6 Ground-State Preparation by Filtering

$$d = \Theta\left(\sqrt{\frac{\alpha_H}{\Delta}}\,\log\frac{1}{\varepsilon}\right), \qquad \alpha_H = \alpha_R^2. \tag{2.6}$$

2.7 Readout and Amplitude Amplification

2.8 Synthesis

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3 The Physics-Informed Effective Hamiltonian Method

This chapter presents the physics-informed effective Hamiltonian method of Wu, Paine, Philip, Gentile and Kyriienko [1] and evaluates it. The method turns a differential equation into a ground-state problem in three steps. The unknown function is expanded in Chebyshev polynomials, and its coefficients are stored as the amplitudes of an n-qubit state $|\psi\rangle$. The equation and each of its boundary or data conditions are then written as matrix operators that annihilate $|\psi\rangle$ when it holds the solution's coefficients: the paper's introductory example $f'' + f = 0$ becomes $(\mathbb{G}^2 + I)\,|\psi\rangle = 0$, where $\mathbb{G}$ differentiates a Chebyshev series. Finally, the Gram matrices of these operators are summed into a positive semi-definite effective Hamiltonian H whose ground state is the solution. Solving the equation thereby becomes preparing a ground state, which needs a block encoding of H, a filter that removes its excited spectrum, and a measurement that reads the field back through the same Chebyshev feature map. The construction descends from differentiable quantum circuits [2] through their Chebyshev latent space [3].

The paper's case for the method is that it needs no variational training, that its solution can be evaluated at any point of the domain rather than on a grid, and that "necessary terms have $O(\mathrm{poly}(n))$ scaling and can be implemented efficiently" [1]. §§ 3.1 to 3.4 present the model, the Hamiltonian, the circuits and the nonlinear extension in the paper's own terms and equations, and § 3.5 assesses them. The construction is stated in full, in the form this thesis runs, in § 4.

3.1 The Chebyshev Latent-Space Model

The model is an overlap [1], [3]. On n qubits, with $N = 2^n$, the paper defines the quantum Chebyshev state or feature map

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}}\, T_0(x)\,|0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}} \sum_{k=1}^{N-1} T_k(x)\,|k\rangle\,, \tag{3.1}$$

whose components are the first N Chebyshev polynomials of the first kind, and writes the solution as the scaled overlap

$$f_Q(x) = \sqrt{\eta}\, f_q(x), \qquad f_q(x) = \langle\tau(x)|\psi\rangle_n\,, \qquad \langle\psi|\psi\rangle = 1, \tag{3.2}$$

where the amplitudes of $|\psi\rangle$ are the series' weighted coefficients and $\eta > 0$ is a scaling factor that carries the norm a unit vector cannot (the paper's Eqs 2–3). The N states $|\tau(\bar{x}_j)\rangle_n$ at the Chebyshev nodes $\bar{x}_j = \cos[\pi(j + \frac{1}{2})/N]$ are orthonormal, but away from the nodes $|\tau(x)\rangle_n$ is not a unit vector. A circuit prepares the normalised state on one ancilla and n system qubits by post-selection, and the paper then discards the norm $\| \, |\tau(x)\rangle_n \, \|$, "since it can be implicitly incorporated into the scaling factor", so that its derivations use the unnormalised

state of Equation (3.1). Every constraint below is linear in $|\psi\rangle$ and fixes only its direction. The scale is recovered after the ground state $|\psi_g\rangle$ has been prepared, from one known value of the solution, the *regular* datum $f(x_s) = y_s$: $\sqrt{\eta} = y_s / \langle \tau(x_s) | \psi_g \rangle_n$ (the paper's Eq. 11).

3.2 Hamiltonian Construction

Differentiation acts on the coefficients by a fixed matrix, $f'_q(x) = \langle \tau(x) | \mathbb{G} | \psi \rangle_n$, with $\mathbb{G}$ the Chebyshev differentiation matrix (the paper's Eq. 4). The paper applies its transpose to a row state and prints it as $\mathbb{G}_n^\top$; this thesis writes the same matrix acting on coefficient vectors. $\mathbb{G}$ is upper triangular and dense, because the derivative of T_p contains every T_k of the opposite parity below it [4]: one high coefficient of f reaches every lower coefficient of f'. Its entries are given in § 4.2. Higher derivatives are its powers, so a homogeneous linear ordinary differential equation (ODE) with constant coefficients, $\sum_j a_j f^{(j)} = 0$, is represented by

$$A_{\text{std}} |\psi\rangle = 0, \qquad A_{\text{std}} = \sum_j a_j \, \mathbb{G}^j \tag{3.3}$$

(the paper's Eq. 15, there of second order). The subscript "std" lables the paper's regime, in which every derivative is represented as power of $\mathbb{G}$; § 4 states it in full as the *standard form*.

A variable coefficient cannot be substituted in the same way, because the Chebyshev coefficients of a product are not the products of the factors' coefficients. The paper uses the lift of Paine et al. [3], $x^p \langle \tau(x) |_n = \langle \tau(x) |_{n+1} \mathbb{M}_{x^p}$ (its Eq. 5), a constant $2N \times N$ matrix that carries a power of x onto a register one qubit wider, so that every term of the equation shares the $(n+1)$-qubit basis. A polynomial coefficient, the only kind the paper's examples use, is a finite sum of such lifts; any other coefficient would be truncated as a Maclaurin series in the same way. A source $r(x)$ does not scale with $|\psi\rangle$ and is brought into the same form through the regular datum: the collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}_n(x_s)/f(x_s)$, with $\mathbb{B}_n$ defined below, maps the solution's coefficients to the constant $1/\sqrt{\eta}$, so a source expanded in a Maclaurin series truncated at order $\bar{p}$ becomes an operator on $|\psi\rangle$ (the paper's Eqs 11–14). The general linear equation $\sum_j a_j(x) f^{(j)} = r(x)$, with coefficients and source expanded in powers of x, is then represented by

$$A_{\text{std}} = \sum_j \mathbb{M}_{a_j} \mathbb{G}^j - \mathbb{M}_r \, \mathbb{D}^{(0)}(x_s), \qquad \mathbb{M}_{a_j} = \sum_p a_{j,p} \, \mathbb{M}_{x^p}, \qquad \mathbb{M}_r = \sum_{p \le \bar{p}} c_p \, \mathbb{M}_{x^p}, \tag{3.4}$$

with $a_{j,p}$ the monomial coefficients of $a_j(x)$ and c_p the Maclaurin coefficients of $r(x)$; the paper's three examples with transcendental sources take $\bar{p} = 3$, 5 and 7.

Boundary and initial conditions enter as data constraints built from the rank-one row $\mathbb{B}_n(x) = \sqrt{2^n} \, |0\rangle^{\otimes n} \langle \tau(x) |_n$ (the paper's Eqs 9–10), which maps the model to the constant function equal to its value at x. An *invariant* constraint has a zero target and is unchanged by scaling: $f(x_z) = 0$ gives $\mathbb{B}_n(x_z) |\psi\rangle = 0$, and $f'(x_m) = 0$ gives $\mathbb{B}_n(x_m) \mathbb{G} |\psi\rangle = 0$. The paper places these at boundaries where the solution vanishes or at interior zeros and extrema of the known solution, printed to two decimal places for the inhomogeneous examples ($x = -0.2$, -0.73, 0.91

and -0.75), and uses one invariant constraint for each second-order ordinary example, with the regular datum fixing the scale.

Two variables are carried on two n-qubit registers with the product state $\langle\tau(x)|_n \otimes \langle\tau(y)|_n$; derivatives act as $\mathbb{G} \otimes I$ and $I \otimes \mathbb{G}$, and a constraint on a boundary as $\mathbb{B}_n(x_z) \otimes I$ (the paper's Eqs 22–28). The invariant sets of the three partial differential equation (PDE) examples are chosen by hand. Laplace's equation uses its three homogeneous sides, and its non-zero side $f(x,1) = \sin(\pi(x+1)/2)$ appears in no constraint. The heat equation $\partial_t f = \frac{1}{25}\partial_x^2 f$ uses $f(t,\pm1) = 0$ and $\partial_x f(t,\pm\frac{3}{4}) = 0$, and the wave equation $\partial_t^2 f = 4\partial_x^2 f$ uses $f(t,\pm1) = 0$, $f(\frac{1}{8},x) = 0$ and $f(\frac{7}{8},x) = 0$. The points $\pm\frac{3}{4}$, $\frac{1}{8}$ and $\frac{7}{8}$ are zeros of the known solutions.

Each operator is turned into a Gram matrix, and the Gram matrices are summed:

$$H_{\text{std}} = \mathcal{T}(A_{\text{std}}) + \sum_i \mathcal{T}(\mathbb{B}_i), \qquad \mathcal{T}(X) = X^\top X, \qquad \langle\psi|H_{\text{std}}|\psi\rangle = \|A_{\text{std}}\psi\|^2 + \sum_i \|\mathbb{B}_i\psi\|^2, \tag{3.5}$$

where $\mathbb{B}_i$ runs over the invariant constraints (the paper's Eq. 16). The energy of a state is its total squared violation of the equation and its data. The paper describes H_{std} as "a real symmetric positive-definite matrix with distinct eigenvalues" and recasts the equation as the eigenvalue problem $H_{\text{std}}\,|\psi_g\rangle = \lambda_{\min}\,|\psi_g\rangle$ with $\lambda_{\min} \approx 0$. It calls H_{std} physics-informed because $\mathcal{T}(A_{\text{std}})$ embeds the differential constraint as the loss of a physics-informed neural network (PINN) embeds it, while the $\mathcal{T}(\mathbb{B}_i)$ embed the data. However nothing is trained, and the ground state is prepared rather than fitted. The paper's worked residual at $N = 4$ and the entries of the lifts are given in Appendix C.

3.3 Circuits, Filtering and Readout

Block encodings. The paper block-encodes each term of H_{std} separately and combines them by a linear combination of unitaries (LCU) (§ 2.4). The rank-one rows $\mathbb{B}_n(x)$ and $\mathbb{D}^{(0)}(x_s)$ pair the inverse Chebyshev feature map with a reflection about the all-zero state in a two-term combination (its Figs 2 and 8), on $n+2$ qubits at subnormalisation $\sqrt{2^{n+1}}$, divided by $f(x_s)$ for $\mathbb{D}^{(0)}$, and run "with high probability". $\mathbb{G}$ is encoded approximately by two conditioned subroutines that load its entries, which grow linearly along each odd superdiagonal, as rotation amplitudes (its Fig. 9). The circuit's top-left block is $\mathbb{G}$ divided by the prefactor

$$\alpha_{\mathbb{G}}^{\text{paper}} = \left(2^{n-1} + 2^n\right)\|\mathbb{G}\|_S, \qquad \|\mathbb{G}\|_S = \max\left(\|\mathbb{G}\mathbb{G}^\top\|_1,\ \|\mathbb{G}^\top\mathbb{G}\|_1\right), \tag{3.6}$$

on $2n+3$ qubits, "with negligible errors in matrix entries that exponentially decrease as n increases", and with $O(n^3)$ gates once the multi-controlled gates are decomposed: "polynomial in n and thus efficient" [1]. The registers, subroutines and angles are given in Appendix D.1.

The imaginary-time filter. The ground state is prepared by quantum imaginary-time evolution (QITE), which multiplies each eigencomponent of an initial state by $e^{-\beta E}$, so that the

lowest dominates once β exceeds the inverse gap. The paper applies it to $\tilde{H} = H_{\text{std}}/\|H_{\text{std}}\|_F$ by an indefinite-parity quantum singular value transformation (QSVT) sequence whose phases are fitted to $e^{-\beta x}$ (§§ 2.5 and 2.6); it writes t for the imaginary time, whereas we use β. The time is set by an "approximate lower bound",

$$\beta \;\geq\; \frac{\lambda_{\max}}{\lambda_2\,(n-1)}, \tag{3.7}$$

with $\lambda_{\max}$ and λ_2 the largest and second-smallest eigenvalues of $\tilde{H}$ (or H_{std}), and two examples are reported to need $\beta = 15$ and 8 with even and odd polynomial degrees $(6, 7)$. For the cost, the paper expects a preparation time of order $\Delta^{-1}\,\text{polylog}(1/\varepsilon)$ by eigenstate filtering, or $\Delta^{-1/2}$ by algorithmic thermalisation, and notes that the gaps "are problem-dependent and in general are not easy to bound". Its phase sets, the dilation of $\tilde{H}$ and the parameters of its examples are given in Appendix K.5.

Readout. The field is read out point by point by an interferometric overlap measurement [5], [6]. Measuring the all-zero outcome after the inverse feature map returns only the squared overlap, so the paper measures two probabilities, one for the prepared state and one for that state superposed with a reference, and combines them into the signed overlap $\langle\tau(x)|\psi_g\rangle$ and, with η, the field value $f_Q(x)$ (its Eqs 36–39). The identity in the form we use is Equation (8.12).

3.4 Doubled-Space Extension to Nonlinearity

A nonlinear term such as $(f')^2$ cannot be expressed as a linear operator on $|\psi\rangle$. The paper's treatment doubles the space: the unknown becomes $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$ on $2n$ qubits, a product of two model functions is the overlap of $|\Psi\rangle$ with a product of two feature states, and the product-to-sum rule folds that product back into one Chebyshev series on one extra qubit,

$$\langle\tau(x)|_n \otimes \langle\tau(x)|_n = \langle\tau(x)|_{n+1}\;\mathtt{m}_1, \qquad \mathtt{m}_1 \in \mathbb{R}^{2N\times N^2} \tag{3.8}$$

(its Eq. 8), so that $f^{(i)}f^{(j)}$ becomes $\mathtt{m}_1(\mathbb{G}^i \otimes \mathbb{G}^j)$ on $|\Psi\rangle$. Linear terms are carried into the same space by tensoring them with the collapse row $\mathbb{D}^{(0)}(x_s)$, which supplies the constant of one copy (its Eq. 30), and the Hamiltonian is built as in Equation (3.5). The paper demonstrates the construction on two quadratic ordinary equations at $n = 2$ and reports that the Hamiltonian then has at least 2^{2n-1} degenerate zero-energy states, so that solving the equation is "converted to that of seeking a degenerate state" that "closely matches the analytical solution". The construction is quadratic: a term of degree p needs p copies, a register of dimension N^p (Appendix H.2), as Paine's concurrent work also observes [7].

3.5 Critical Assessment

Strengths. The construction is unified: ordinary and partial differential equations and their data constraints take one least-squares form, and nonlinear terms have a route into it. Its

kernel is exact whenever the solution is a polynomial within the truncation. The Chebyshev basis needs no spatial grid, so the model and all its derivatives are available at every $x \in [-1, 1]$ from one prepared state, and the Hamiltonian has few terms, each with polynomially many gates. It also provides a non-variational solution pathway.

Weaknesses in the cost. The most serious weakness is that the spectrum of the Hamiltonian, not its circuits, may fix the cost. A polynomial filter must suppress the excited spectrum relative to the ground state, and its degree grows with the square root of the ratio of the encoding's subnormalisation, which is at least $\|H_{\text{std}}\|$ for any block encoding, to the gap (Equation (2.6)). The derivatives enter H_{std} through powers of $\mathbb{G}$, whose norm grows as N^2 (§ 4.2), and H_{std} is a Gram product, the normal-equations form of a least-squares problem, which squares the conditioning of the operator it minimises [8]. Where the gap does not grow with N, the norm, and with it the cost, therefore grows with every derivative the equation carries, whatever the circuit; the paper's gate counts are the cost of one query, and the number of queries is set by that norm. The expected time $O(\Delta^{-1}\,\text{polylog}(1/\varepsilon))$ leaves its units open: if Δ is the gap of $\tilde{H}$, it is the gap of H_{std} divided by $\|H_{\text{std}}\|_F$, and the norm re-enters there. A second weakness sits on top of the first: the prefactor of Equation (3.6) is not compared with $\|\mathbb{G}\|_2$, which it exceeds by a factor that grows with N, so the circuit for $\mathbb{G}$ is loose even against the matrix it encodes. § 4.5.1 and § 4.6 separate the two.

Weaknesses in the evidence. The paper's examples support the method less than they appear to. Each inhomogeneous example combines a rounded constraint point with a truncated Maclaurin source, and either alone sets an error floor that no increase in n removes: a zero imposed where the solution does not vanish contradicts the equation, and a truncated source poses a different equation (§§ 4.2 and 4.3, Appendix K). The PDE examples take their invariant points from the known solution, which a new problem does not supply, and their invariant sets omit the initial or boundary function that fixes the solution, so the kernel degenerates as n grows beyond the printed resolution. For the heat equation the natural completion, its initial function at $t = 0$, lies inside the time domain $[-1, 1]$ and would pose a backward, ill-posed problem on $[-1, 0]$ (§ 4.3). The description of H_{std} as positive definite with distinct eigenvalues is stronger than a sum of Gram matrices allows, and the case the method aims at, a solution represented exactly, is a zero eigenvalue (§ 4.4). The time rule (Equation (3.7)) contains no ε, so it cannot certify a target accuracy, and it needs $\lambda_{\max}$ and λ_2, which the method does not supply in advance (Appendix K.4).

Weaknesses in scope. The nonlinear route projects rather than prepares: any function of the Hamiltonian acts as a scalar on its zero-energy eigenspace, so a filter returns the projection of the initial state onto a kernel of dimension at least 2^{2n-1}, and the direction it picks within that kernel is set by the initial state, not by the equation (§ 7.2). The readout identity treats the prepared state as the ground state with certainty, although a filter followed by post-selection succeeds with probability below one, and it returns one point of the field per measurement

campaign (§ 8.6). The construction is stated for one real scalar field, and the paper sets complex-valued states aside, since with a Fourier basis "the resulting total energy function is of a complicated form, rendering it difficult to find the ground state" (§ 6).

Open problems. The paper's Discussion names two: "studying the spectral gap dependence is an important question for the future work", and the choice of basis, for which "one guiding principle here can be the cost of differentiation, where the goal is minimizing the depth of quantum circuits to implement derivatives, multiplication etc." [1]. Paine's concurrent work [7] compares alternative Hamiltonian constructions and function encodings and finds, on a few small instances, that the normalised gap depends on both, while stating that these results do not establish a general ordering. It gives neither the gap's scaling with resolution nor block-encoding circuits, so it raises the first problem without answering it. The first weakness above is the quantitative form of that problem: the gap must be set against the Hamiltonian's norm, and the norm is set by the differentiation matrix, which is the paper's second problem.

Each of the following three chapters centre around a core question:

1. *Does the method's total number of queries originate in the particular encoder circuits, or in the Hamiltonian itself?*
2. *Can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and what does that cost in register size and spectral gap?*
3. *Can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel?*

Together they ask where the norm of the Hamiltonian, and with it the filter degree, comes from, and whether changing what the residual carries can lower it.

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4 The Standard Form and Where Its Cost Lies

This chapter answers the first question of § 3.5: does the method's total number of queries originate in the particular encoder circuits, or in the Hamiltonian itself? The two are separable because a filter's degree depends on the encoding only through its subnormalisation, $d = \Theta\big(\sqrt{\alpha_H/\Delta}\,\log(1/\varepsilon)\big)$ (Equation (2.6)), and every block encoding of H has $\alpha_H \geq \|H\|$. We call the ratio $\alpha_H/\|H\| \geq 1$ the *encoder's cost* and $\|H\|$ itself the *Hamiltonian's cost*: the first belongs to the circuit, the second to the operator. Separating them needs the operator stated exactly. The construction is that of Wu et al. [1], built on the Chebyshev latent space and lifts of Paine et al. [3]. We state it once, in the form the rest of the thesis runs, which we call the *standard form*; where it departs from the printed method we give the reason at the point of departure, and Table 2 collects the departures. §§ 4.1 to 4.4 build the standard form, § 4.5 encodes its derivative exactly, and § 4.6 shows what no encoding removes.

4.1 The Model

We use the paper's model, Equations (3.1) and (3.2), unchanged. Its weights define a normalised basis, $\phi_0 = T_0/\sqrt{N}$ and $\phi_k = T_k/\sqrt{N/2}$ for $1 \leq k < N$, in which $\langle\tau(x)|$ is the row $(\phi_0(x), \ldots, \phi_{N-1}(x))$ and a truncated series $f_N = \sum_k b_k\phi_k$ is the overlap $\langle\tau(x)|b\rangle$, with $|\psi\rangle = b/\|b\|$. With $x = \cos\theta$ the Chebyshev weight turns $\int T_jT_k(1-x^2)^{-1/2}\,dx$ into $\int_0^\pi \cos j\theta\cos k\theta\,d\theta$, so the basis is orthogonal with the common squared norm π/N, and Parseval's identity gives the ideal value of the paper's scale factor,

$$\eta_e = \|b\|^2 = \frac{N}{\pi}\int_{-1}^{1}\frac{f_N(x)^2}{\sqrt{1-x^2}}\,dx. \tag{4.1}$$

It is fixed by the solution and the resolution alone and grows like N for a smooth solution, and it is the value the regular datum returns, to within the truncation's error at x_s, when the prepared state is $b/\|b\|$. The datum also fixes the sign: $\sqrt{\eta} = y_s/\langle\tau(x_s)|\psi_g\rangle$ is negative when the ground state, which is defined only up to sign, has the opposite sign to the datum, so the paper's $\eta > 0$ is a convention and the decoded field carries the sign.

The feature vector is kept unnormalised, as the paper's derivations use it. Its squared norm, $\big(1 + 2\sum_{k\geq1} T_k(x)^2\big)/N$, oscillates about one and reaches $(2N-1)/N$ at $x = \pm1$, and every constraint row of § 4.3 inherits that weight. Normalising $\tau(x)$ point by point would change the rows' relative weights, and with them the least-squares solution, so the circuits load the unnormalised vector. A function of several variables is carried on one register per axis, each with its own n, with the feature vector $\tau(x)\otimes\tau(y)$ in Kronecker order. The running example of this chapter is the paper's Fig. 3b, the critically damped oscillator $f'' + 4f' + 4f = 0$ with $f(-1) = 0$ and $f(0) = \frac{1}{2}$, whose solution is $f = \frac{1}{2}(1+x)e^{-2x}$; in the benchmark set of Table 7

it is R2a.

4.2 Differentiation and Multiplication

In the normalised basis the differentiation matrix has the entries

$$[\mathbb{G}]_{k,p} = \begin{cases} 2p, & p > k \geq 1,\ p-k \text{ odd}, \\ \sqrt{2}\,p, & k = 0,\ p \text{ odd}, \\ 0, & \text{otherwise}, \end{cases} \tag{4.2}$$

and it is exact on the represented space, since the derivative of a polynomial of degree below N stays in it [4]; the paper prints its transpose (§ 3.2). $\mathbb{G}$ is strictly upper triangular and nilpotent, $\ker \mathbb{G}^j$ is the space of polynomials of degree below j, and about half the positions above the diagonal are non-zero. Its norm grows as N^2: $\|\mathbb{G}\|_2/N^2$ converges to $1/(2j_{-3/4,1}) \approx 0.472$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and $\|\mathbb{G}^k\|_2/N^{2k}$ converges to a positive constant for every k. Because the basis is orthonormal at the Chebyshev nodes, $\mathbb{G}$ is orthogonally similar to the matrix that differentiates nodal values, so the growth is not an artefact of the normalisation. The structure and the bounds are proved in Appendix B.3, and the limits in Appendix E.3.

A polynomial coefficient $a(x)$ acts by the lift of Paine et al. [3], $x^p \,\langle \tau(x)|_n = \langle \tau(x)|_{n+1}\, \mathbb{M}_{x^p}$ (the paper's Eq. 5), extended by linearity to $\mathbb{M}_a = \sum_p a_p \mathbb{M}_{x^p}$. The lift is exact whenever $\deg a \leq N$, because the product of a series of degree below N with a then has degree below $2N$ and fits the wider register without truncation. A constant enters as $a_0\mathbb{M}_1 = a_0\sqrt{2}\,[\,I;\,0\,]$, the $\sqrt{2}$ being the ratio of the normalisations on $n+1$ and n qubits, and $\|\mathbb{M}_a\|_2 \leq \sqrt{2}\sup|a|$. The standard form keeps the paper's lift; the descriptor form of § 5 uses the square Galerkin truncation instead. The equation $\sum_j a_j(x) f^{(j)} = r(x)$ then has the equation block

$$A_{\text{std}} = \sum_j \mathbb{M}_{a_j}\mathbb{G}^j - \mathbb{M}_r\,\mathbb{D}^{(0)}(x_s), \qquad \mathbb{M}_r\,\mathbb{D}^{(0)}(x_s) = \frac{|\hat{r}\rangle\,\langle \tau(x_s)|}{f(x_s)}, \tag{4.3}$$

in which the source enters through the collapse row of § 4.3, with $\hat{r}$ the coefficients of r on the lifted register.

The paper expands a transcendental source in a Maclaurin series truncated at order $\bar{p}$ (Equation (3.4)). A truncated source poses a different equation, whose solution the ground state then approximates. The order-7 series of the source $x\,e^{-2x}\sin 4x$ of the paper's Fig. 5d (R4d, $f'' + 4f' + 20f = x\,e^{-2x}\sin 4x$) differs from the source by more than the source itself at $x = -1$, and the resulting error is a floor that no increase in n removes. The standard form loads the Chebyshev coefficients of the source instead, through the same collapse row, so the source is as accurate as the series that represents the solution.

The paper treats at most two variables. On any number of axes a term is the Kronecker

product over the axes of $\mathbb{M}_a\mathbb{G}^{j_a}$, each axis lifted if any term on it needs the exact product, and a constant coefficient on a lifted axis enters through $\mathbb{M}_1$ (Appendix C.2). The benchmark problems use at most two.

4.3 Constraints and Data

The constraints are built from the paper's rank-one rows (its Eqs 9–11). The zero row $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\,\langle\tau(x)|$ maps a series to the constant function $f_N(x)$, so $\mathbb{B}(x)\mathbb{G}^j$ vanishes exactly on the series whose j-th derivative vanishes at x. The collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$ maps the solution to the constant 1 whatever its scale, which is what lets a source enter Equation (4.3). Both are homogeneous in the scale, so neither fixes it; the regular datum does, after preparation (§ 4.1).

A zero row is exact only at a point where the solution's derivative of that order vanishes. The paper's inhomogeneous examples (its Fig. 5) place their invariant constraints at zeros and extrema of the known solution printed to two decimal places: R4a, $(x-1)f''-xf'+f = (x-1)^2$, imposes $f'(-0.2) = 0$ for an extremum at $x \approx -0.19598$. At a rounded point the row demands a zero the solution does not have, so least squares trades the equation against the constraint and the error floors. On R4a, whose source is a polynomial and so represented exactly, the error stays at 3.1×10^{-3} for every n from 4 to 10, against 3.8×10^{-14} at $n = 4$ with the exact point. The standard form takes the point the problem defines, the zero or extremum the paper rounded, exactly. That point is problem data, inherited from the paper's statement, not something the method computes: a new problem must supply its constraint points with its data. On R4b to R4d the rounded point and the truncated source each floor the error on their own, and the error converges only with both replaced (the triage is in Appendix K.2).

How many constraints a problem needs is a property of the continuous equation. For a homogeneous equation of order k whose leading coefficient has no zero on $[-1, 1]$, the solutions form a k-dimensional space S, and those that satisfy the invariant constraints form a subspace of dimension $k - r$, where r is the rank of the constraints restricted to S. The solution is unique up to the scale the regular datum fixes exactly when $r = k-1$: fewer leave a degenerate kernel, and $r = k$ leaves only $f = 0$ (Proposition 8). The rank is that on S, not that of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. Whether the truncated residual then has a one-dimensional kernel with a usable gap at a given n is decided by computing it, and so is uniqueness for an inhomogeneous equation or a leading coefficient that vanishes (the trusted kernel, § 8.2).

A partial differential equation has an infinite-dimensional solution space, and the count has no analogue. The paper's invariant sets (§ 3.2) leave open what fixes the solution. On Laplace's equation (R5a) the modes $\sin\frac{\ell\pi(x+1)}{2}\,\sinh\frac{\ell\pi(y+1)}{2}$, $\ell \geq 1$, all satisfy the equation and its three homogeneous sides, and only $\ell = 1$ matches the omitted top side, which reaches the model through the regular datum alone. At the printed resolution the truncation excludes the others;

as n grows they become representable, and the kernel grows to dimension unknown at $n = (5, 5)$. The sets of the heat equation (R5b) and the wave equation (R5c) degenerate in the same way at larger n, and their interior points are read off the known solutions. The standard form imposes the initial or boundary function itself, on its slice. For a function g on the slice $x_a = x^*$ the rows are

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle \tau(x_s)|}{f(x_s)}, \tag{4.4}$$

with $\mathbb{S}_a(x^*)$ the zero row $\mathbb{B}(x^*)$ on axis a and the identity on the others, and $\hat{g}$ the coefficients of g broadcast along axis a. On a multiple cf of the solution they return $c\,(f|_{x_a=x^*} - g)$, so they are homogeneous in the scale and vanish exactly on the solution and its multiples (Appendix C.3). These datum slices need nothing but the problem's data and serve both forms, and the kernel no longer degenerates as n grows: once the resolution represents the solution, it is one-dimensional and trusted at every larger n computed, to $n = (6, 6)$. The heat equation's natural datum, its initial function at $t = 0$, lies inside the time domain $[-1, 1]$, and imposing it there would pose a backward heat problem on $[-1, 0]$, which is ill posed; the standard form imposes the initial function at $t = -1$, chosen so that the solution is the paper's, and the problem runs forward.

The paper's associated Legendre examples R3c and R3d (its Figs 4c and 4d, $l = 5$ and 6 with $m = 1$) multiply the equation through by $1 - x^2$ to clear its $m^2/(1 - x^2)$ term. That leaves the leading coefficient $(1 - x^2)^2$, with a double zero at each end, and a solution $P_l^1 = -\sqrt{1 - x^2}\, P_l'$ with a branch point at each end, which a Chebyshev series approximates only algebraically: at the printed $n = 5$ neither problem has a trusted ground state, and the error on R3c is 8.7×10^{-1}. The standard form solves for $v = f/\sqrt{1 - x^2}$ instead, whose equation $(1 - x^2)v'' - 4xv' + \big(l(l + 1) - 2\big)v = 0$ has polynomial coefficients, a leading coefficient with simple zeros and a polynomial solution, so the series converges spectrally, to 4.9×10^{-15} on R3c (Appendix C.4). It is also the form the descriptor can run (§ 5.6).

Two smaller points complete the data. The paper prints the solution of its Laplace example as $\operatorname{csch}\pi\, \cosh(\pi x/2) \sinh(\pi(y+1)/2)$, which does not satisfy its boundary conditions; the solution, and the reference used throughout, is $\cos(\pi x/2) \sinh(\pi(y + 1)/2)/\sinh\pi$. Vector data, such as the initial values of a coupled system, enter as ratio rows, defined with systems in § 6.2.

4.4 The Standard Form

The equation block and the constraint rows are stacked into one residual,

$$R_{\text{std}} = \begin{pmatrix} A_{\text{std}} \\ C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}} = A_{\text{std}}^\dagger A_{\text{std}} + \sum_i C_i^\dagger C_i, \tag{4.5}$$

every block at weight one, so that H_{std} is the paper's sum of Gram matrices, Equation (3.5), with the datum slices among the C_i. The ground state of H_{std} is the smallest right singular

vector of R_{std}. The classical reference computes it from R_{std} by a singular value decomposition, or by Lanczos iteration beyond its reach (§ 8.2), and never forms H_{std}, which would square the residual's condition number: on R2a at $n = 8$ the relative gap of H_{std} is 2.3×10^{-17}, below double precision. The circuit likewise encodes R_{std}, not H_{std} (§ 8.3).

Written this way, H_{std} is positive semi-definite, and in the very case the method aims at, a solution the truncation represents exactly, it is not definite, contrary to the paper's description (§ 3.2). When the solution is a polynomial of degree below N, $\mathbb{G}$ differentiates it and the lifts multiply it exactly, so its coefficients are annihilated by R_{std} and give H_{std} an exact zero eigenvalue; Legendre's equation $(1 - x^2)f'' - 2xf' + 20f = 0$ (R3a, the paper's Fig. 4a), whose solution is P_4, has one from $n = 3$. Otherwise the truncated solution leaves a residual, $\lambda_0 > 0$, and the ground state minimises the total squared residual: the construction is least squares. Nor need the bottom of the spectrum be distinct, as the paper's partial differential sets show. The thesis relies on neither property; it relies on an isolated ground state, measured as a trusted kernel (§ 8.2).

Definition 10 (The standard form): For a linear problem on one or more Chebyshev axes, the *standard form* is the stacked residual R_{std} of Equation (4.5), whose equation block is Equation (4.3), lifted on every axis where a coefficient or a source needs the exact product, and whose constraint rows are of four kinds, each at weight one: zero rows $\mathbb{B}(x^*)\mathbb{G}^j$ at a point or on a slice; the collapse row $\mathbb{D}^{(0)}(x_s)$, through which sources enter; ratio rows for vector data (§ 6.2); and datum slices (Equation (4.4)) for initial and boundary functions. Its Hamiltonian is $H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}}$, and its solution is the smallest right singular vector of R_{std}, scaled by the regular datum $f(x_s) = y_s$.

Table 6 lists the four kinds as model inputs, and a benchmark label such as R2a names the problem with the data of this section. Table 2 collects every departure from the printed method: those argued above, and those of the pipeline, argued where they are built. Each changes the data a residual carries, or how the residual is solved, encoded, filtered or read, and none changes $\mathbb{G}$, the lifts or the Gram structure. The standard form therefore has the paper's Hamiltonian norm to leading order, and with it the paper's cost, and is only more accurate, so a comparison against it is conservative. The one exception is R3c and R3d, whose substitution changes the equation itself; their cost is compared in the form the standard form runs.

The departures are checked against the paper itself. With every printed choice restored, the rounded points, the Maclaurin sources and the hand-picked invariant sets, the same construction reproduces the matrices printed in the paper's supplementary material at $n = 2$ entry by entry, and its basis gives, for the exact solution, the printed scale factor of each of the paper's eleven ordinary examples to within one unit of the last printed digit: for Fig. 3b, as printed at $n = 3$, the exact solution gives $\eta_e = 1.29009$ against the printed 1.29. The printed scale factors of its three partial differential and two nonlinear examples are not reproduced, by the exact solution's η_e or by any ground state we computed, and the cause is not settled. The record of every example, the triage of the printed choices and the verdict on each of the paper's claims

Table 2: **Every departure from the printed method removes a stated defect or adds a stated capability.** The method of Wu et al. [1], as printed, against the standard form and pipeline of this thesis. Rows above the rule change the residual; rows below it change how the residual is encoded, solved, filtered or read. None changes the differentiation matrix $\mathbb{G}$, the lifts or the Gram structure of the Hamiltonian.

Aspect	Wu et al.	This thesis	Defect removed or property gained	Where
Constraint points	zeros and extrema of the known solution, rounded (-0.2, -0.73, 0.91, -0.75)	the problem's point, exact	a zero the solution lacks contradicts the equation; the error floor is removed	§ 4.3
Sources	Maclaurin series to order $\bar{p} = 3, 5, 7$	Chebyshev coefficients on the lifted register	a truncated source poses another equation; spectral accuracy	§ 4.2
PDE data	invariant sets chosen by hand; the top side and the initial functions omitted	datum slices for initial and boundary functions	no point read off the solution; no degeneracy as n grows; the heat problem posed forward	§ 4.3
Double zero (R3c, R3d)	the equation times $1 - x^2$	solved for $v = f/\sqrt{1 - x^2}$	simple zeros and a polynomial solution; spectral convergence	§ 4.3
Laplace reference	$\operatorname{csch} \pi \cosh(\pi x/2) \sinh(\pi(y+1)/2)$	$\cos(\pi x/2) \sinh(\pi(y+1)/2)/\sinh \pi$	a typo	§ 4.3
Axes	two	any number (two run)	generality	Appendix C.2
Gram sum	$\sum \mathcal{T}(\cdot)$	the stacked residual R, H never formed	the same H; R solved and encoded without squaring its conditioning	§ 4.4
Fields	one real scalar field	several fields; ratio rows for vector data; complex fields by realification	coupled systems and the Carleman lift	§ 6
Nonlinearity	the doubled space	the doubled space, and a Carleman lift	a preparation, not a projection, for polynomial systems	§ 7
Encoding of $\mathbb{G}$	rotation-loaded, approximate; α looser than $\|\mathbb{G}\|_2$ by about N^3	an exact factorisation; α within 2.12 of $\|\mathbb{G}\|_2$	the encoder's cost removed for constant coefficients	§ 4.5
Normalisation	$H/\|H\|_F$	H/α_R^2 of the stacked encoding	a block encoding's own subnormalisation	§ 8.3
Filter	imaginary time by mixed-parity QSVT, phases fitted, time by Equation (3.7)	minimax edge filter by generalised quantum signal processing (GQSP), phases by Weiss's complement and the inverse nonlinear Fourier transform (NLFT), degree a priori	no parity split; suppression to ε by construction, which the time rule does not give	§ 8.4
Initial state	not specified	geometric product state on the field block; its overlap reported	answer-independent	§ 8.4
Classical solve	the ground state as an eigenvector of H	the smallest right singular vector of R	resolves a gap below double precision on H	§ 8.2
Readout	interferometric, point by point	the field decoded classically from the prepared state	the protocol built at small n only, with a correction for $p_G < 1$	§ 8.6

are given in Appendix K.

4.5 Encoding the Derivative

Every term of the standard form is a product of lifts, rank-one rows and powers of $\mathbb{G}$. The rank-one rows have exact circuits whose subnormalisation is fixed by their own norm, so the term whose encoding decides the encoder's cost is $\mathbb{G}$.

4.5.1 The Paper's Encoding

The paper's circuit for $\mathbb{G}$ (its Fig. 9; § 3.3) selects each entry (k, p) by a register subtraction over a superposition of offsets and loads the column value $2p$ as an amplitude through a ladder of small controlled rotations. Its claims about gates and entry errors hold, the first as a bound and the second to leading order (Appendix D.1). The claim that decides its cost is the prefactor of Equation (3.6), which is valid but not compared with $\|\mathbb{G}\|_2$. Since $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1$ grows as $N^4/3$, the prefactor grows as $N^5/2$, against $\|\mathbb{G}\|_2 \sim N^2/(2j_{-3/4,1})$, so its looseness tends to $j_{-3/4,1}N^3$ with $j_{-3/4,1} \approx 1.06$; at $n = 7$ it is 2.2×10^6 (exact classical computation, Appendix E.4). A circuit with polynomially many gates is cheap to apply once, but the filter applies it a number of times set by its subnormalisation (Equation (2.6)): the encoding is efficient in gates and inefficient in subnormalisation. To cost it like the others, we rebuild its mechanism with registers of our own, the offsets restricted to odd values, since an even offset would load values onto entries of $\mathbb{G}$ that vanish. The rebuilt circuit's own subnormalisation is a third of the printed prefactor, $N\|\mathbb{G}\|_S/2$ against $3N\|\mathbb{G}\|_S/2$, so every cost this thesis quotes for the paper's encoding is the kinder reading. Its entries are approximate, with an error of order N^{-5} (Proposition 9), so no state is prepared with it; it enters the resource estimates only, composed into the same stacked residual as the exact encoding below. The rebuilt circuit, its error bound and the verdict on each of the paper's circuit claims are given in Appendix D.1.

4.5.2 An Exact Factorisation

The paper's circuit is approximate, and its prefactor large, because it loads the column value of each entry as a rotation amplitude. The structure of $\mathbb{G}$ allows a different route. Every non-zero entry of Equation (4.2) is a product of three factors, each depending on one index alone: a row weight, equal to $1/\sqrt{2}$ in row 0 because of the normalisation of T_0 and to one elsewhere; a selector depending only on the offset $p - k$, non-zero when $p > k$ and $p - k$ is odd; and the column value $2p$, the weight with which differentiation passes T_p to the lower polynomials. Each factor has an exact block encoding with no amplitude loaded by rotation.

Proposition 1 (Exact factorisation of the differentiation matrix)**.** *For every $n \geq 1$ the differentiation matrix on the normalised Chebyshev coefficients factorises as*

$$\begin{aligned} \mathbb{G} &= R_0\left(2U_{\text{odd}}\right)\Lambda, \\ R_0 = I - \left(1 - \tfrac{1}{\sqrt{2}}\right)|0\rangle\langle 0|, \qquad \left(U_{\text{odd}}\right)_{kp} &= \left[p > k,\ p - k\ \textit{odd}\right], \qquad \Lambda = \operatorname{diag}(p), \end{aligned} \tag{4.6}$$

where U_{odd} is the sum of the $N/2$ truncated shifts by odd offsets.

The proof compares the two sides entry by entry with Equation (4.2) and is given in Appendix E.1. As a check on our implementation, not on the proposition, the block of the composed circuit matches the independently assembled $\mathbb{G}$ entrywise to 3.4×10^{-13} (absolute) for $n \leq 7$.

4.5.3 Circuits and Cost

Each factor is encoded as a linear combination of unitaries [9] whose terms are unitary already, so no part of the encoding is approximate. The column index is a sum of Pauli strings,

$$\Lambda = \frac{N-1}{2} I - \sum_{j=0}^{n-1} 2^{j-1} Z_j, \tag{4.7}$$

since $p = \sum_j 2^j(1-z_j)/2$ in terms of the eigenvalues z_j of Z_j. Its $n+1$ terms need $\lceil \log_2(n+1) \rceil$ selector qubits and $O(n)$ gates, and its subnormalisation $\alpha_\Lambda = N-1$ equals $\|\Lambda\|_2$. The row weight is the two-term combination $R_0 = aI + b\,(I - 2\,|0\rangle\langle 0|)$ with $a = \frac{1}{2}(1 + 1/\sqrt{2})$ and $b = \frac{1}{2}(1 - 1/\sqrt{2})$, both positive, so $\alpha_{R_0} = 1 = \|R_0\|_2$.

The sum of shifts is the one factor that needs arithmetic (Figure 1). Its PREPARE puts an offset register in the uniform superposition over the $N/2$ odd offsets, by an X on the lowest qubit and Hadamards on the rest. Its SELECT is the in-place subtraction $p \leftarrow p-d$ of a ripple-carry circuit [10], which sets a borrow flag when $p < d$. The terms that would wrap around carry a borrow of one, and post-selecting the flag on zero removes them exactly. The block is therefore $\frac{2}{N}\big[p > k,\ p-k \text{ odd}\big]$, an exact encoding of U_{odd} at $\alpha = N/2$ with $O(n)$ Toffoli gates, and one PREPARE serves all $N/2$ terms because SELECT is a permutation (Appendix E.2).

Quantikz circuit: offset register d (X on the lowest qubit, $H^{\otimes(n-1)}$ on the rest), system register p, carry and borrow ancilla; a ripple-carry subtractor $p \leftarrow p - d$ spanning all five wires; PREPARE†; post-selection of d and the borrow flag on $\langle 0|$.

Figure 1: **The odd-offset shifts are one subtraction.** PREPARE (left) puts the offset register d in a uniform superposition over the $N/2$ odd offsets, by an X on its lowest qubit and Hadamards on the rest. SELECT is a ripple-carry subtractor [10] that replaces the system register p by $p - d$ and sets the borrow flag when $p < d$. PREPARE† (right), with post-selection of d and the borrow flag on $|0\rangle$, keeps $\frac{2}{N}\,|k\rangle\langle p|$ for every odd $d = p - k \geq 1$, and the wrapped terms vanish exactly. The block is $U_{\text{odd}}/(N/2)$, so $\alpha = N/2$, on n system and $n+2$ ancilla qubits.

The three encodings multiply, and so do their subnormalisations. Their product is an exact block encoding of $\mathbb{G}$ at $\alpha_{\mathbb{G}} = 1 \cdot N \cdot (N-1)$, with the following cost.

Proposition 2 (Cost of the structured encoding)**.** *The factorisation of Proposition 1 compiles to an exact block encoding of* $\mathbb{G}$ *on* n *system qubits with*

$$\alpha_{\mathbb{G}} = N(N-1), \qquad \frac{\alpha_{\mathbb{G}}}{\|\mathbb{G}\|_2} \longrightarrow 2j_{-3/4,1} \approx 2.117, \qquad O(n^2) \text{ gates}, \qquad n+4 \text{ ancilla}, \tag{4.8}$$

of which $n-1$ *ancilla are the selector of the* $N/2$*-term combination* U_{odd}*.*

The limit is that of $\|\mathbb{G}\|_2/N^2$ (§ 4.2); the gate count assumes an ancilla-free synthesis of multi-controlled gates, and falls to $O(n \log n)$ with a linear-cost synthesis that borrows idle qubits [11]. The proof is given in Appendix E.3. The ancilla count is exact in the circuit for $n = 2$ to 8. The transpiled two-qubit gate counts are available only at small n (109 at $n = 2$ and 471 at $n = 5$), consistent with the bound but too few to fit it. The limit is asymptotic, and the sizes we reach are measured: the ratio $\alpha_{\mathbb{G}}/\|\mathbb{G}\|_2$ is 1.62 at $n = 2$ and rises to 2.11 at $n = 10$ (Table 3), against a ratio for the paper's encoding that grows as N^3. The structured encoding is this close to the norm because α_{R_0} and α_Λ equal their factors' norms. Its looseness comes from U_{odd}, whose α exceeds its norm by at most $\pi/2$, and from the product, in which the large entries of Λ meet the columns of U_{odd} with few non-zero entries.

The $n-1$ selector qubits are not an artefact of this circuit. A combination of L terms needs $\lceil \log_2 L \rceil$ selector qubits, and lower bounds of this kind hold for exact block encodings built from L-term decompositions [12], so the count is the price of encoding the $N/2$ shifts of U_{odd} as one combination. It marks a structural difference between the two forms: the descriptor form of § 5.7 never forms the sum of shifts, and its ancilla grow as $\log n$.

Table 3: **The structured encoding stays within a constant of the norm; the paper's prefactor does not.** Subnormalisation of the two encodings of the differentiation matrix, relative to $\|\mathbb{G}\|_2$. The structured encoding of Proposition 2 rises towards its limit $2j_{-3/4,1} \approx 2.117$; the paper's prefactor (Equation (3.6)) grows as N^3. The full table to $n = 10$ is Table 15. Exact classical computation of the norms and subnormalisations.

n	$\|\mathbb{G}\|_2$	Structured (Proposition 2)	The paper's (Equation (3.6))
2	7.39	1.623	49.9
4	121	1.986	4.05×10^3
7	7739	2.100	2.20×10^6
10	495 308	2.115	1.14×10^9

The remaining terms of the residual are built from the same pieces. A coefficient $a(x)$ is encoded through its lift as a sum of powers of the single-x lift, itself a combination of truncated shifts built by the same post-selected subtraction, so its subnormalisation is the sum of the absolute monomial coefficients weighted by powers of that lift's (Appendix C.7). The rank-one rows $\mathbb{B}(x)$ and $\mathbb{D}^{(0)}(x_s)$ are the paper's exact circuits, and a source given by its Chebyshev coefficients is loaded by one rotation tree. A power $\mathbb{G}^k$ is a product of encodings that share

one ancilla register, with a failure flag per product, so $\mathbb{G}^2$ costs one qubit more than $\mathbb{G}$. The subnormalisation of a product is the product of the subnormalisations, so a power compounds the constant above while $\|\mathbb{G}^k\|_2 < \|\mathbb{G}\|_2^k$. The stacked residual is then assembled into a single reflection (§ 8.3), and the composed circuit is verified against the independently assembled Hamiltonian (§ 8.5).

4.6 What Remains: The Standard-Form Hamiltonian

For constant coefficients the structured encoding removes the encoder's cost from the whole residual, not only from $\mathbb{G}$. On R2a at $n = 8$ the structured encoding of the stacked residual has $\alpha_R = 4.26 \times 10^9$ against $\|R_{\text{std}}\|_2 = 5.51 \times 10^8$, a ratio of 7.74 (a resource estimate; the circuit at this size is not simulated). The ratio rises from 4.72 at $n = 2$ with increments that roughly halve at each added qubit, so it is levelling off over $n = 2$ to 8, although we have not proved a limit for the whole residual. The Hamiltonian is encoded at $\alpha_H = \alpha_R^2$, so $\alpha_H/\|H_{\text{std}}\|$ is the square of that ratio, from 22.3 to 59.9, whereas the paper's encoding of $\mathbb{G}$ alone is loose by a factor growing as N^3.

No encoder can change the norm it encodes. For an equation of order k with constant coefficients, a non-zero leading coefficient and constraints of lower order, $\|\mathbb{G}^k\|_2/N^{2k}$ tends to a positive constant, so $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$ and $\|H_{\text{std}}\| = \|R_{\text{std}}\|_2^2 = \Theta(N^{4k})$ (Appendix E.3). The factor N^2 per derivative is the norm of $\mathbb{G}$, and the exponent doubles because the Hamiltonian is the Gram product $R^\dagger R$. The gap, by contrast, is measured, not proved. On R2a and on $f'' - 2f' - 3f = 0$ (R2b, the paper's Fig. 3c) it does not grow with N: on R2a $\Delta = 7.04$ at $n = 7$ and 7.04 at $n = 8$, while $\|H_{\text{std}}\|$ grows from 1.2×10^{15} to 3.0×10^{17}. Since $\alpha_H \geq \|H_{\text{std}}\|$ for every block encoding, Equation (2.6) gives

$$\|H_{\text{std}}\| = \Theta\big(N^{4k}\big), \qquad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Theta}\big(N^{2k}\big), \tag{4.9}$$

which at second order, the order of the paper's examples, is $\|H_{\text{std}}\| = \Theta(N^8)$ and $d = \tilde{\Theta}(N^4)$ whatever the block encoding. This is the Hamiltonian's cost. The degree at the ideal-encoding bound $\alpha = \|R\|_2$, which contains no encoding, grows as N^{2k} on the constant-coefficient one-axis problems measured, from $n = 7$ to 8: as $N^{\boxed{\text{unknown}}}$ on the first-order R1 and as $N^{4.00}$ on R2a, as Equation (4.9) predicts with a flat gap. The floor is stated for the minimax filter that prepares every state in this thesis; for polynomial filters in general it is what the optimality of Lin and Tong's construction supports [13], and we claim no more. At $n = 8$ the estimated degree for R2a is 1.2×10^{10} as built and 1.5×10^9 at the ideal-encoding bound, both resource estimates far beyond what exact emulation reaches. The degree is linear in α_R, so the ratio of the two is the encoder's looseness and the second is the Hamiltonian's share.

The polynomial gate count that Wu et al. report for their operators [1], and that the rebuilt circuits bear out (Appendix D.1), is the cost of one query to the block encoding. The cost of the method is that cost times the number of queries, and Equation (4.9) fixes the number of

queries by the norm, however few gates a query takes. The paper does not claim otherwise: it leaves the gap, and so the preparation time, to future work. Its claim of efficiency is therefore true of the circuit and silent on the method, and polynomial gates per query do not make the standard form efficient, since its filter makes $\tilde{\Theta}(N^{2k})$ queries or more, which is exponential in n.

The growth of the degree between $n = 7$ and $n = 8$, again from resource estimates, shows the two costs apart. With the paper's encoding the degree grows as $N^{10.0}$, with the structured encoding as $N^{4.01}$, and at the ideal bound as $N^{4.00}$. The first difference is the encoder's cost, which the structured encoding removes; the common exponent is the Hamiltonian's. It belongs to the least-squares functional, not to the circuits: $H_{\text{std}} = R^\dagger R$ is the normal-equations form of the least-squares problem for an operator of order k, and such a problem squares the conditioning of the operator it minimises. The classical remedies are to recast the equation as a first-order system [8] or to use a basis in which differentiation is banded (§§ 2.3.2 and 2.3.3).

The conclusion has a limited scope. The structured encoding bounds $\alpha_H/\|H_{\text{std}}\|$ for constant coefficients only, and is loose for a variable coefficient that is lifted. On R3a the circuit pays for the product $\mathbb{M}_a\mathbb{G}^2$ as $\|\mathbb{M}_a\|\,\|\mathbb{G}^2\| \gg \|\mathbb{M}_a\mathbb{G}^2\|$, because the coefficient $1 - x^2$ vanishes at the ends of the interval, where the derivatives of Chebyshev polynomials are largest. The measured norm of this residual grows as N^2, not as $N^{2k} = N^4$, and the encoder's cost reappears: the as-built degree grows as $N^{4.01}$, while at the ideal bound it grows as $N^{2.00}$, the exponent the descriptor's degree also has there ($N^{2.00}$ at the ideal bound). At $n = 8$ the as-built degree is 1.7×10^{10} against 1.1×10^5 at the ideal bound, both resource estimates. For this reason § 9.3 quotes the ideal-encoding bound beside every as-built cost, and where the comparison between the forms reverses at that bound, as it does on R3, reports the reversal as a finding.

This answers the first question. For the constant-coefficient problems of the paper's regime the cost of the method has two parts: the encoder's, a factor growing as N^3 in the subnormalisation of the paper's encoding of $\mathbb{G}$, which the structured encoding removes; and the Hamiltonian's, $\tilde{\Theta}(N^{2k})$ in the filter's degree, which no encoding removes. § 9.3 tests both on the benchmark problems. What remains open to change is the Hamiltonian itself, through what its residual carries, and § 5 changes it by carrying the derivatives as unknowns instead of eliminating them into the dense $\mathbb{G}$.

remove once previous section complete

5 The Descriptor Reformulation

This chapter answers the second question of § 3.5: can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and at what cost? We build the reformulated residual, the descriptor form, and prove that its kernel is the standard form's (§§ 5.1 to 5.4). We then identify the difficulty it introduces, a spectral gap eroded by the carried derivatives, and the a priori rescaling that restores it on one axis (§§ 5.5 and 5.6). Finally we construct its block encoding and derive what it costs and what it does to the filter's degree (§§ 5.7 to 5.9).

5.1 The Idea

§ 4 located the cost of the standard form in its Hamiltonian: $\|\mathbb{G}^k\|_2 = \Theta(N^{2k})$, so $\|H_{\text{std}}\| = \Theta(N^{4k})$ whatever the encoder. This density comes from requiring the derivative's coefficients be expressed in the same basis and the function's. If the derivative may instead live in a neighbouring polynomial basis, differentiation is banded, and $\mathbb{G}$ factors into two banded matrices whose product is dense only because one of them is inverted.

The density of the standard form belongs to its representation, not to differentiation. Differentiation is a dense, upper-triangular matrix on Chebyshev coefficients only because the derivative is required to be expanded in the same basis as the function. Classical spectral methods avoid this by working in the *integral form* of the Galerkin (or tau) method [14]: the highest derivative is taken as the unknown, lower derivatives and the function are recovered from it by integration, and the equation is written over these quantities. Integration maps each Chebyshev polynomial to a combination of its two neighbours, so the integration operator is banded where the differentiation operator is dense, and the discretised equation inherits the sparsity. Equivalently, and in the form we use, each derivative is expanded in the neighbouring ultraspherical basis, in which differentiation is banded [4]. In either description $\mathbb{G}$ factors into banded matrices, and its product is dense only because one of the factors is inverted. We therefore change the form in which the equation is written and leave the equation, its boundary data and its solution as they were.

The descriptor form keeps the factors instead of the product. Where the standard form substitutes $\mathbb{G}^j$ for each derivative, the descriptor form carries $f', \dots, f^{(k)}$ as unknowns beside f, ties consecutive derivatives by banded recurrences, and writes the equation over all of them. A first-order equation or system needs no extra unknowns, because the equation itself gives the derivative: multiplying the standard rows through by one banded factor is enough. We call the first layout the *chain* and the second the *mass form* (§ 5.3).

Neither ingredient is new to numerical analysis. Banded differentiation is the ultraspherical

spectral method of Olver and Townsend [4], of which the factors of § 5.2 are the discrete shadow (§ 2.3.2). Carrying derivatives as unknowns so as not to square the condition number of a least-squares functional is the idea of first-order-system least squares [8] (§ 2.3.3). The physics-informed Hamiltonian is a least-squares functional, so both apply to it. The contribution of this thesis is their transfer to a block-encoded physics-informed Hamiltonian: a proof that the kernel is unchanged, an encoding of the residual as one flat linear combination of unitaries, an a priori rescaling that restores the gap the new unknowns erode, and a measurement of what all of this does to ground-state preparation.

Figure 2 shows the effect on the running example $f''+4f'+4f = 0$ (R2a). At $n = 7$ the standard form's stacked residual holds 8384 non-zero entries on 128 unknowns, and the descriptor's holds 1402 on 512: four times as many unknowns, because of the block register that § 5.3 counts, and six times fewer entries. That trade, a larger state for a sparser residual, is the subject of the chapter. Carrying the intermediate derivatives instead of eliminating them turns the residual from dense to banded with its kernel unchanged, so that the Hamiltonian's norm falls from $\Theta(N^{4k})$, which is $\Theta(N^8)$ at second order, to $\Theta(N^2)$ at fixed block weights; the price is a block register and a gap that the derivative blocks erode, which an a priori rescaling restores on one axis at no gate cost.

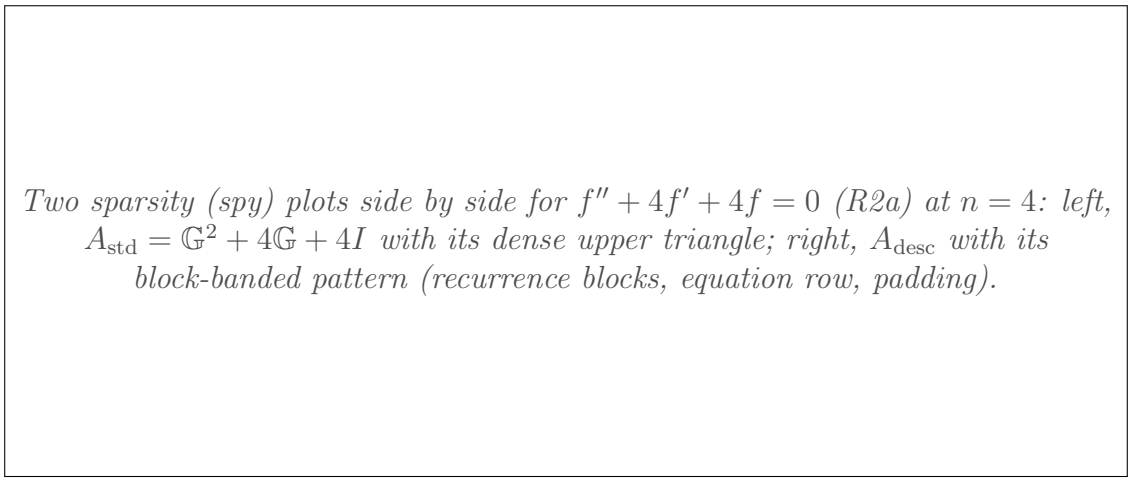


Figure 2: **Dense against banded: the two residuals of the same equation.** Sparsity of the residual operators of $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$. Left, the standard form's $A_{\text{std}} = \mathbb{G}^2 + 4\mathbb{G} + 4I$, whose upper triangle is full. Right, the descriptor's A_{desc}, whose recurrence blocks, equation row and padding are banded. The descriptor's matrix is four times the size and, at $n = 7$, holds six times fewer non-zero entries (§ 5.1).

5.2 The Banded Factorisation

Let $f = \sum_\ell a_\ell T_\ell$ have derivative $f' = \sum_\ell p_\ell T_\ell$. The two coefficient sequences are tied by the recurrence

$$c_{\ell-1}\, p_{\ell-1} - p_{\ell+1} = 2\ell\, a_\ell, \qquad \ell \geq 1, \tag{5.1}$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$, the factor on p_0 being an artefact of the normalisation of T_0 (the derivation is given in Appendix B). Each row ties one coefficient of the function to two of

the derivative. Truncated to N coefficients and written in the normalised basis of § 3.1, with $\hat{f}^{(0)}$ and $\hat{f}^{(1)}$ the coefficients of f and f', it is the matrix identity

$$\tilde{\mathbb{B}}\,\hat{f}^{(1)} = \tilde{\mathbb{D}}\,\hat{f}^{(0)}. \tag{5.2}$$

$\tilde{\mathbb{D}}$ has a single non-zero diagonal, the first superdiagonal, with entries $(2\sqrt{2}, 4, 6, \dots, 2(N-1))$; $\tilde{\mathbb{B}}$ has the diagonal $(2, 1, 1, \dots)$ and the second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries against the $\Theta(N^2)$ of $\mathbb{G}$. Two properties are used throughout: $\tilde{\mathbb{B}}$ is upper triangular with a non-zero diagonal, so it is invertible; and since the singular values of a single-diagonal matrix are its entries' magnitudes, $\|\tilde{\mathbb{D}}\|_2 = 2(N-1)$, while $\|\tilde{\mathbb{B}}\|_2 \le 2 + \sqrt{2}$.

The dense differentiation matrix is therefore the quotient

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{5.3}$$

exactly at every N. The inverse $\tilde{\mathbb{B}}^{-1}$ is a triangle of ones on each parity class, and the density of $\mathbb{G}$ is that inverse; Equation (5.2) is the same linear map before the inverse is taken. This is the discrete form of the ultraspherical spectral method [4]: up to the basis normalisation, $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ are twice that method's operators for conversion from first-kind to second-kind Chebyshev polynomials and for differentiation into the second kind (Appendix B). The ultraspherical method applies these operators to eliminate the derivatives and obtain a banded system in f alone. We instead keep Equation (5.2) as a row of the system and the derivative as an unknown, which lets one construction serve every order, every axis and every coupled field.

5.3 Two Layouts and Kernel Equivalence

Consider the linear ODE of order k

$$\sum_{j=0}^{k} a_j(x)\, f^{(j)}(x) = r(x), \qquad x \in [-1, 1], \tag{5.4}$$

with the data constraints of § 3.2, every function in it represented by its first N coefficients in the normalised Chebyshev basis. Which layout carries it depends on its order.

The chain ($k \ge 2$). Let $\hat{f}^{(j)} \in \mathbb{C}^N$ be the coefficient vector of $f^{(j)}$ for $j = 0, \dots, k$, stacked into the state

$$w = (\hat{f}^{(0)};\, \hat{f}^{(1)};\, \dots;\, \hat{f}^{(k)}). \tag{5.5}$$

Two kinds of row couple the blocks. The *recurrence rows*

$$-\tilde{\mathbb{D}}\,\hat{f}^{(j)} + \tilde{\mathbb{B}}\,\hat{f}^{(j+1)} = 0, \qquad j = 0, \dots, k-1, \tag{5.6}$$

apply Equation (5.2) block by block. They contain no derivative operator, and on the kernel they are equivalent to $\hat{f}^{(j+1)} = \mathbb{G}\hat{f}^{(j)}$. The *equation row*

$$\sum_{j=0}^{k} \mathbb{M}_{a_j}\, \hat{f}^{(j)} - \mathbb{M}_r\, \mathbb{D}^{(0)}(x_s)\, \hat{f}^{(0)} = 0 \tag{5.7}$$

is the equation itself, with $\mathbb{M}_{a_j}$ the multiplication operator of § 3.2 (it reduces to $a_j I$ for a constant coefficient, and lifts the row to $n+1$ qubits wherever the standard form lifts it) and the source entering as it does there. The derivatives are unknowns of w and are never replaced by $\mathbb{G}^j \hat{f}^{(0)}$, so the dense quotient Equation (5.3) never appears. The name is borrowed from descriptor systems in control theory, which likewise keep intermediate quantities as unknowns beside the state and tie them by algebraic rows.

Definition 11 (Descriptor residual operator): Collecting the recurrence rows and the equation row, the descriptor residual operator A_{desc} is the block matrix

$$A_{\text{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & & \\ & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ \mathbb{M}_{a_0} - \mathbb{M}_r\mathbb{D}^{(0)} & \mathbb{M}_{a_1} & \cdots & \cdots & \mathbb{M}_{a_k} \end{pmatrix} \tag{5.8}$$

with $(k+1)N$ columns, so that $A_{\text{desc}}\, w = 0$ is the equation. Its first k block rows are the recurrences and its last is the equation row.

The padded block register. On a quantum register the block index is held by $n_f = \lceil \log_2(k+1) \rceil$ qubits, which address 2^{n_f} blocks rather than $k+1$. The state is therefore padded to $2^{n_f} N$ amplitudes by blocks $b > k$, each pinned to zero by its own row $c_{\text{pad}} \hat{f}^{(b)} = 0$ with $c_{\text{pad}} = 1$. No other row reads a padding block, so the padding decouples: the Hamiltonian is block diagonal, and the spectrum of the physical blocks is unaffected. The padding is inert but not free. The chain's state has $2^{n_f} N$ amplitudes against the standard form's N, which is n_f more qubits and up to twice the unpadded $(k+1)N$ when $k+1$ is not a power of two: a second-order equation carries f, f' and f'' in four blocks. The constraint rows of § 5.4 are added to these block rows.

The mass form ($k = 1$). A first-order equation or system $z' = \mathbb{A}z$ with C components has in the standard form the rows $I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N$, one block per component. The equation already determines each derivative, so a derivative block would carry nothing new; the mass form instead multiplies the rows on the left by $I_C \otimes \tilde{\mathbb{B}}$ and uses $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$,

$$(I_C \otimes \tilde{\mathbb{B}})\big(I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N\big) = I_C \otimes \tilde{\mathbb{D}} - \mathbb{A} \otimes \tilde{\mathbb{B}}. \tag{5.9}$$

Both factors are banded and neither contains $\mathbb{G}$. The name is taken from the ODE literature, $\tilde{\mathbb{B}}$ playing the mass matrix of $\tilde{\mathbb{B}}\dot{z} = \dots$. For $f' + 2f = 0$ the mass form is $\tilde{\mathbb{D}} + 2\tilde{\mathbb{B}}$ on N unknowns, where a chain would need $2N$ unknowns and elimination through $\mathbb{G}$ gives the dense $\mathbb{G} + 2I$; for a scalar equation with a constant coefficient it is the ultraspherical method's own operator. The mass form has the standard form's state, initial state and readout, and differs from it only by a banded invertible factor on the equation rows. It adds no qubits and has no derivative block to rescale (§ 5.6). Since the Carleman lifts of § 7 and the coupled fields of § 5.4 are first-order systems, the mass form is the layout the nonlinear problems use.

Both layouts rest on one fact.

Theorem 1 (Kernel equivalence)**.** *Let R_{std} stack A_{std} with the constraint rows of § 3.2. Let R_{desc} stack the descriptor's recurrence, equation and padding rows with the same constraints, each reading the block of the derivative it constrains (§ 5.4), and let R_{mass} be the mass form's stacked residual. Then*

$$\ker R_{\text{desc}} = \left\{ (p,\ \mathbb{G}p,\ \dots,\ \mathbb{G}^k p,\ 0)\ :\ p \in \ker R_{\text{std}} \right\}, \qquad \ker R_{\text{mass}} = \ker R_{\text{std}}. \tag{5.10}$$

Proof. Write $Ep = (p; \mathbb{G}p; \dots; \mathbb{G}^k p; 0)$. Chain: the padding rows force the padding blocks of a kernel vector w to zero, and since $\tilde{\mathbb{B}}$ is invertible each recurrence row gives $w_{j+1} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}\, w_j = \mathbb{G}w_j$, so $w = Ew_0$. The recurrence rows vanish on Ep for every p, because $(\tilde{\mathbb{B}}\mathbb{G} - \tilde{\mathbb{D}})\mathbb{G}^j p = 0$, and every other row of R_{desc} acts on Ep as the standard form's row of the same name acts on p: the equation row substitutes $\mathbb{G}^j p$ for $\hat{f}^{(j)}$, and a constraint row reads the block that holds $\mathbb{G}^j p$. Hence $R_{\text{desc}}Ep$ vanishes exactly when $R_{\text{std}}p$ does, and E is injective because its first block is the identity. Mass form: R_{mass} is R_{std} with its equation rows multiplied on the left by the invertible $I_C \otimes \tilde{\mathbb{B}}$ (Equation (5.9)) and its constraint rows unchanged. □

The theorem concerns kernels, and it is exact. An exact kernel exists only where the solution lies in the span of the basis; otherwise both residuals have full rank, and their ground states are the smallest right singular vectors of different matrices. The descriptor measures a state's residual against a norm that carries its derivatives (§ 5.5), and the mass form weights the equation rows by $\tilde{\mathbb{B}}$. The two ground states converge to the same function as n grows, at rates we measure but do not prove: on R2a at $n = 7$ the decoded field's error against the analytic solution is 8.9×10^{-15} in the descriptor and 2.8×10^{-11} in the standard form (§ 9.4). We therefore claim equal kernels and observe a common limit, not equal ground states.

5.4 Constraints and Partial Differential Equations

A data constraint sits on the block that holds the quantity it constrains. A Dirichlet condition $f(x_z) = 0$ reads block 0 with the point-evaluation row $\mathbb{B}(x_z)$, as in the standard form. A Neumann condition $f'(x_m) = 0$ reads block 1 with the same row, $\mathbb{B}(x_m)\hat{f}^{(1)} = 0$, where the standard form must write $\mathbb{B}(x_m)\mathbb{G}\,|\psi\rangle$. The regular datum of § 3.2 reads block 0, so the field

is decoded and scaled exactly as in the standard form.

On several axes the field is block 0, and each axis a owns a chain $\partial_a f, \dots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it. The recurrences use that axis's factors $\tilde{\mathbb{B}}_a, \tilde{\mathbb{D}}_a$, tensored with the identity on the other axes. The heat equation ($k_t = 1$, $k_x = 2$) carries four blocks, and Laplace's and the wave equation ($k = 2$ on each axis) five, padded to eight. A mixed derivative would need a block of its own; the construction does not build one, which limits it to equations without mixed derivatives. § 6.1 treats coupled fields. Initial and boundary *functions*, as opposed to point values, enter both forms as datum slices, rows that pin a slice of the state to a known function. Datum slices are available to the standard form too, and neither form doubles the state of a linear PDE, so the descriptor's advantage on these problems is not one of state size.

5.5 The Gap of a Carried Derivative

Carrying the derivatives costs the descriptor something the standard form does not pay. With E the map of Theorem 1, the descriptor's Rayleigh quotient on the states that satisfy the recurrences is

$$\frac{\|R_{\text{desc}} E p\|^2}{\|E p\|^2} = \frac{\|R_{\text{std}}\, p\|^2}{\sum_{m=0}^{k} \|\mathbb{G}^m p\|^2}, \tag{5.11}$$

because the recurrence rows vanish on Ep and the numerator is the standard form's. By the min–max principle the descriptor's singular values are bounded above by the generalised singular values of this pair. The denominator carries every derivative, so the gap is flat in N only where the equation controls each carried derivative: where no state p orthogonal to the solution has a small residual against a large derivative. This is an upper bound, not a proof of flatness, and we measure the gap instead. The standard form measures the same states by $\|p\|$, against which their residuals are large, so its gap stays flat; its difficulty is the other side of the quotient, the norm of Equation (4.9).

Try to prove that the gap is flat for a regular one-axis equation (the lower bound on the quotient of

For a one-axis equation whose leading coefficient has no zero on $[-1, 1]$, the continuum a priori estimate $\|f^{(k)}\| \le (\|Lf\| + \|\text{lower-order terms}\|)/\min|a_k|$ suggests that the quotient stays bounded away from zero on the complement of the kernel, and the measured gap is indeed flat: on R2a the descriptor's gap is 2.87 at $n = 7$ and 2.87 at $n = 8$. It fails in three ways.

1. *A leading coefficient that vanishes.* A polynomial of degree N can concentrate within $\mathcal{O}(N^{-2})$ of an endpoint, where the Chebyshev points cluster, and each of its derivatives grows by a factor $\mathcal{O}(N^2)$ there. Where a_k vanishes at that endpoint, the leading term $a_k p^{(k)}$ of such a state is small, the lower-order terms are small as well, and the quotient is small against the top block. Without rescaling the gap falls as N^{-4} at a simple zero at an endpoint and as N^{-2} at an interior one; on the Legendre equation R3a, whose leading coefficient is $1 - x^2$, it is 2.7×10^{-8} at $n = 8$.

2. *Two axes.* The equation constrains only a sum of top derivatives, $f_{xx} + f_{yy}$ or $f_t - \kappa f_{xx}$.

A high-frequency state with $\mathbb{G}_x^2 p \approx -\mathbb{G}_y^2 p$, or with its weight in the block the equation weights least, has a small residual against large blocks, and only the datum rows penalise it. On the heat equation $\partial_t f - \frac{1}{25}\partial_x^2 f = 0$ (R5b, the paper's Fig. 6d) the unscaled gap is 2.5×10^{-7} at $n = 5$ per axis.

3. *The field block crowded out.* The ground state's weight on the field block is about $\|b\|/(\sum_m \|\mathbb{G}^m b\|^2)^{1/2}$, with b the solution's coefficients, and the growth of the derivatives makes it small. An initial state prepared on the field block then overlaps the ground state by at most that weight, and amplification costs its reciprocal. On the stiff equation $f'' + 5f' + 400f = 0$ (R2c, the paper's Fig. 3d) the unscaled gap is 7.5×10^{-6} and the field block's weight 5.4×10^{-6} at $n = 7$.

The collapse is not an artefact of the lift, the padding, the arithmetic precision, the constraints, the basis or the classical reference: each was tested and ruled out, and a descriptor built in the ultraspherical basis collapses at the same rates where the leading coefficient vanishes. The tests and the full mechanism are given in Appendix G.

5.6 Block Rescaling

The remedy is to rescale the carried derivatives so that no block outweighs the equation's own scale. The block of order m is carried divided by ζ^m,

$$\hat{f}^{(m)} \longmapsto \zeta^{-m}\,\hat{f}^{(m)}, \qquad m = 0, \ldots, k, \tag{5.12}$$

which replaces R_{desc} by $R_{\text{desc}}S$ with $S = \operatorname{diag}(1, \zeta, \ldots, \zeta^k) \otimes I_N$. Because S is diagonal and invertible, the kernel becomes $S^{-1} \ker R_{\text{desc}}$ and Theorem 1 survives. The field block has weight one, so the readout is unchanged, and the padding keeps its weight and still decouples. What changes is the Hamiltonian, $SR^\dagger RS$. This is a congruence, not a similarity, so the spectrum moves: by Equation (5.11) the denominator becomes $\sum_m \zeta^{-2m}\|\mathbb{G}^m p\|^2$, and ζ sets how much each carried derivative weighs against the residual.

In the circuit the rescaling costs nothing. Scaling the columns of block m by ζ^m multiplies the coefficient of every term of the block encoding of § 5.7 by ζ raised to the order of its column block, which adds no gate and no ancilla and changes only the subnormalisation. A general diagonal equilibration, with a different weight for each amplitude, would need a diagonal block encoding of its own at $\Theta(\dim w)$ gates; that is why the weights are constant on each block, and general equilibration appears in no circuit in this thesis.

The a priori rule. ζ is computed from the equation's coefficients and the resolution, never from a solution or a spectrum. For a one-axis equation whose leading coefficient a_k has no zero on $[-1, 1]$,

$$\zeta = 2\left(\bar{a}_j/\bar{a}_k\right)^{1/(k-j)}, \tag{5.13}$$

with j the lowest order present and $\bar{a}$ the mean of $|a|$ over the interval. For constant coefficients and $j = 0$, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the magnitudes of the characteristic roots: the scale at which a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m \|f\|$, so that dividing block m by its m-th power balances the blocks at the equation's own scale. Where a_k has a zero on the interval, $\zeta = N/2$, which makes the quotient of the concentrated states of § 5.5 of order one. On several axes $\zeta_a = N_a$ per axis, balanced across axes by the top-order coefficients, so that the wave equation $f_{tt} = c^2 f_{xx}$ takes (cN, N). The two constants, the 2 in Equation (5.13) and the $\frac{1}{2}$ in $N/2$, were fixed once by sweeping ζ over the regular and the singular benchmark problems, and are never refitted to a problem; the rule is then checked against the sweep's best ζ on every problem. The filter's cost stays within twice its least value over a window of ζ two to eight times wide, so ζ need only be right to about a factor of two.

What it restores. On every one-axis problem the descriptor runs, the rescaled gap no longer falls with N, and the field block regains its weight. The gap meant here is that of the physical subspace, the one the filter meets, since the padding alone would cap the gap of the whole spectrum at one. On R2c the gap rises from 7.5×10^{-6} to 0.715 and the field block's weight from 5.4×10^{-6} to 0.774 at $n = 7$. On R3a the rescaled gap is 2.27 at $n = 5$ and 3.62 at $n = 8$, against 2.7×10^{-8} unscaled: it rises slowly with n rather than falling.

What it does not remove. Three limits remain, each a finding.

1. *The norm grows with ζ.* Block m carries ζ^m, so $\|R\| \gtrsim \zeta^k \bar{a}_k$ and $\|H\| \gtrsim \zeta^{2k}$. With ζ constant this costs nothing. With $\zeta \propto N$, which a simple zero requires, $\|H\| = \Theta(N^{2k})$, and at a flat gap the degree at second order is about $\tilde{\Theta}(N^2)$, against the standard form's $\tilde{\Theta}(N^4)$ (§ 5.9).

2. *Two axes.* The rescaled gap still falls with N, though far more slowly than unscaled: on R5b it is 0.253 at $n = 3$ per axis and 0.0942 at $n = 5$.

3. *A double zero is beyond any block rescaling.* If a_k vanishes to second order at an endpoint, the concentrated states have quotient $\mathcal{O}(N^{-8}\zeta^{2k})$, which is the unscaled N^{-8} at $\zeta = 1$, so balancing them needs $\zeta \propto N^{4/k}$. At $k = 2$ that is $\zeta \propto N^2$, at which $\|H\| = \Theta(N^8)$, the standard form's norm. The paper's own Legendre panels of Figs. 4c and 4d (R3c and R3d) multiply the associated Legendre operator by $1 - x^2$ and so have exactly this double zero. The descriptor instead runs the equivalent problem for $v = f/\sqrt{1 - x^2}$, whose leading coefficient has simple zeros, and we make no claim for the double zero.

Block rescaling is the descriptor's counterpart of the diagonal preconditioner of the ultraspherical method [4], which bounds the condition number of its banded system. Here the diagonal scales the state's columns and is constant on each block, which is why it costs nothing in the circuit.

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5.7 The Block Encoding: One Flat Linear Combination

Every block of A_{desc} is a short sum of *bands*, each a diagonal followed by a truncated shift. The diagonal of $\tilde{\mathbb{D}}$ is $2c$ on column c, affine in the index except at column 1, where the normalisation of T_0 intervenes; the diagonals of $\tilde{\mathbb{B}}$ and of a constant multiplication are constant except at their ends; and a variable coefficient's multiplication and the Galerkin truncation contribute a few further exceptions. An affine diagonal is a sum of $n+1$ Pauli-Z terms, and each exception is one reflection Q_m:

$$\operatorname{diag}(c)_{c=0}^{N-1} = \sum_{j=0}^{n-1} 2^j \, \frac{I - Z_j}{2}, \qquad Q_m = I - 2\,|m\rangle\langle m|\,. \tag{5.14}$$

A_{desc} is therefore one flat LCU of $\mathcal{O}(n)$ terms,

$$A_{\text{desc}} = \sum_{t=1}^{L} c_t \, U_t, \qquad U_t = (\text{field word}) \otimes \text{SHIFT}^{s_t} \, P_t, \qquad \alpha_A = \sum_{t=1}^{L} |c_t|, \tag{5.15}$$

where the field word acts on the block register, SHIFT^{s_t} is the truncated shift by s_t, and P_t is a Z-string or a reflection. An exactly affine diagonal whose end values share a sign is encoded tightly, at $\alpha = \max_c |a + bc|$, so the dominant term $\tilde{\mathbb{D}}$ is encoded at its own norm $2(N-1)$ and its exceptions cost $\mathcal{O}(1)$. A truncated shift of the N-dimensional register is not unitary; it is applied on the register widened by one overflow qubit, so that rows falling outside the interval land in the half that the final projection discards. That qubit is the shift's only ancilla.

Algorithm 1 compiles the terms into the circuit of Figure 3.

Quantikz circuit: selector registers (PREPARE loads $\sqrt{|c_t|/\alpha_A}$ over their joint index), field register (n_f qubits), system register (n qubits) widened by one overflow qubit; SELECT applies each term's field word, diagonal (Z-string or reflection) and truncated shift (a ladder of multi-controlled-X gates); PREPARE†; post-selection on the selectors, the overflow and the flag.

Figure 3: **The whole descriptor residual is one PREPARE, one SELECT and one PREPARE†.** PREPARE loads $\sqrt{|c_t|/\alpha_A}$ over the joint index of the factored selector registers. SELECT applies each term's field word on the block register (n_f qubits), then on the system register (n qubits, widened by one overflow qubit) its diagonal, a Z-string or a reflection, and its truncated shift. Post-selecting the selectors, the overflow qubit and the flag on $|0\rangle$ leaves A_{desc}/α_A, with the block rescaling carried entirely in the coefficients c_t.

Input: the blocks of A_{desc} as bands, the block weights ζ
Output: PREPARE and SELECT circuits; the subnormalisation α_A
Write every block as a sum of bands: a diagonal and a truncated shift;
Fit each diagonal as affine in the column index plus a few exceptions, Equation (5.14);
Merge terms with equal axis factors across blocks; expand the resulting block-register pattern in Pauli words;
Multiply the coefficient of each term by ζ^m for its column block m;
$\alpha_A \leftarrow \sum_t |c_t|$; PREPARE loads $\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ on the joint selector index of term t;
SELECT: controlled on each selector register, apply the field word, then on each axis the diagonal and the shift;
return *PREPARE, SELECT,* α_A;

Algorithm 1: Compiling the rows of A_{desc} into PREPARE and SELECT.

The terms (lines 1–4). Lines 1 and 2 are the decomposition above. Line 3 gathers terms whose axis factors agree across the rows and columns they join, so that the block register carries their pattern once, as a small matrix expanded in Pauli words; the recurrences' common $\tilde{\mathbb{D}}$ is paid for once and not k times. Line 4 is the whole of the block rescaling.

PREPARE and SELECT (lines 5–6). The selectors are factored: one register for the field operation and, on each axis, one for the diagonal and one for the shift, each only as wide as the number of distinct values its terms use. SELECT then applies each distinct operation once, controlled on its own register, rather than once per term, which is what keeps the ancilla at $\mathcal{O}(\log n)$ (§ 5.8). The rank-one rows, the sources and the data constraints, are encoded through the feature map, with a rotation tree loading a source's coefficients, so no dense state preparation appears anywhere. The constructions and the term-by-term accounting are given in Appendix G.4.

5.8 Cost of the Descriptor Encoding

Proposition 3 (Cost of the descriptor encoding)**.** *For a descriptor residual of a fixed problem, the flat LCU of Equation* (5.15) *is an exact block encoding of the system rows at* $\alpha_A = \sum_t |c_t|$*, with* $\mathcal{O}(\log n)$ *ancilla and* $\mathcal{O}(n^2)$ *gates. For the second-order family R2,*

$$\lceil \log_2(n+4) \rceil + 8 \text{ ancilla}, \qquad \mathcal{O}(n^2) \text{ gates}. \tag{5.16}$$

On a regular one-axis equation at fixed ζ*, the subnormalisation of the system rows is* $\alpha_A / \|A_{\text{desc}}\|_2 = 1 + \mathcal{O}(1/N)$.

The ancilla are the selector registers, the overflow qubit and the flags that project the field terms. Only the diagonal selector's width depends on n: for R2 it indexes the identity, the n gates Z_j and three reflections, so it has $\lceil \log_2(n+4) \rceil$ qubits, while the field and shift selectors have widths fixed by the problem. The gate count follows because each distinct operation is applied once and the shift's multi-controlled ladder costs $\mathcal{O}(n^2)$. The proof is given in Appendix G.5, and the ancilla count is verified for R2's system rows at $n = 2$ to 12.

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The ancilla count is of the least order among exact encodings of this kind. A combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [12], and an affine diagonal has exactly $n+1$ non-zero Walsh coefficients, so any LCU that expands its diagonals in Pauli strings needs $\Omega(\log n)$ of them. We do not claim optimality against every encoding. The structured encoding of the standard form needs $n-1$ selector qubits for $\mathbb{G}$ alone, because U_{odd} is a sum of $N/2$ shifts (§ 4.5.3), so its ancilla grow as $\Omega(n)$ where the descriptor's grow as $\log n$: the difference between a dense operator and a banded one. The better scaling is not yet visible at the sizes that circuits reach. For the whole reflection on R2a the descriptor needs 12 ancilla at $n = 2$ and 13 at $n = 5$, against 10 and 13 for the standard form.

The ratio $\alpha/\|\cdot\|$ is the factor by which an encoding exceeds the least possible subnormalisation, and the filter's degree is linear in it. The proposition concerns the system rows; for the whole stacked residual, constraints included, the ratio is measured. It approaches one on regular one-axis problems, $\alpha_R/\|R\|_2 = 1.04$ on R2a at $n = 8$ (a resource estimate). It stays near two where the leading coefficient vanishes, 2.12 on R3a at $n = 7$, because $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2 a_2$ are then comparable and comparable terms add in ℓ^1 even when each is tight. On two axes it is 2.29 on R5b at $n = 4$ per axis. The limit of one therefore holds for regular one-axis problems only. For comparison, the structured standard-form encoding is at 7.74 on R2a and 3.7×10^4 on R3a, the latter the lifted multiplier's looseness of § 4.6.

The gates per query go the other way on the constant-coefficient problems. On R2a, at the sizes reached, one query of the descriptor costs more than one of the structured standard form, having more terms and a block register: the transpiled two-qubit count of the reflection is 25 914 against 8094 at $n = 2$ and 34 098 against 22 856 at $n = 5$. The ratio falls with n, but both counts are $\mathcal{O}(n^2)$, so the data do not show that it crosses one. The descriptor's advantage is therefore the number of queries, the filter's degree, and not the gates per query, and its chain state is larger by the block register. The next section is where the advantage lies.

5.9 Consequences for the Spectrum and the Filter

Every block of the descriptor is $\mathcal{O}(N)$ at fixed ζ, the recurrences' $\zeta^j\tilde{\mathbb{D}}$ being the largest, so $\|R_{\text{desc}}S\| = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$ whatever the order k: the derivative is banded, and its norm is that of $\tilde{\mathbb{D}}$ rather than of $\mathbb{G}^k$. With the encoding at the norm, $\alpha_H/\|H\|$ falls from 59.9 for the standard form on R2a at $n = 8$ to 1.09 for the descriptor. Since the degree is $d = \Theta(\sqrt{\alpha_H/\Delta}\log 1/\varepsilon)$ (Equation (2.6)), the rescaled gap, which is flat on a one-axis ODE whose leading coefficient does not vanish, gives for that class of problems

$$\|H_{\text{desc}}\| = \Theta(N^2), \qquad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Theta}(N), \tag{5.17}$$

against $\tilde{\Theta}(N^{2k})$ for the standard form with the best exact encoding (Equation (4.9)). Where the leading coefficient vanishes, or on two axes, the rescaling's weights raise the norm and the degree is about $\tilde{\Theta}(N^2)$ or faster; for a double zero it is the standard form's or worse (Table 4).

Neither form is polylogarithmic in N: $\tilde{\Theta}(N)$ is polynomial in N and so exponential in n, and the claim is a lower exponent, not an exponential speed-up. The degree is that of the minimax edge filter, which prepares every comparison in this thesis; for polynomial filters in general the floor is no stronger than Lin and Tong's optimality supports [13].

Table 4: **The descriptor lowers the degree's exponent where the equation controls every carried derivative.** Norm, gap and filter degree of each form by class of problem, with the block weight ζ set by the a priori rule of § 5.6. Norms follow from the block weights and are proved; gaps are measured.

Case	ζ	$\|H\|$	Gap	Degree
Standard, order k, constant coefficients	–	$\Theta(N^{4k})$	flat	$\tilde{\Theta}(N^{2k})$
Descriptor, one axis, $a_k \neq 0$ on $[-1,1]$	fixed	$\Theta(N^2)$	flat	$\tilde{\Theta}(N)$
Descriptor, one axis, simple zero, $k=2$	$N/2$	$\Theta(N^4)$	flat	$\tilde{\Theta}(N^2)$
Descriptor, two axes	N_a per axis	$\Theta(N^4)$	falls	above $\tilde{\Theta}(N^2)$
Descriptor, double zero	$\propto N^2$	$\Theta(N^8)$	–	standard's or worse

On R2a at $n = 8$ the estimated degree is 9.1×10^3 for the descriptor, against 1.2×10^{10} for the standard form as built and 1.5×10^9 at its ideal-encoding bound, all three resource estimates; the measured growth with N is reported in § 9.3. Every comparison is quoted both as built and at $\alpha = \|R\|_2$, which contains neither form's encoding, and where the order reverses, as it does on R3 at the ideal bound and on the heat equation at small n, the reversal is reported as a finding (§§ 9.3 and 9.4).

Two further facts bound what the comparison says. The first is the relative gap: on R2a at $n = 7$ it is $\Delta/\|H_{\text{std}}\| = 5.9 \times 10^{-15}$ for the standard form and 2.8×10^{-6} for the descriptor. An eigensolver applied to H_{std} resolves its ground state only to about machine precision divided by that ratio, and at these sizes it fails, whereas the singular value decomposition of R, whose relative scale is $\Theta(N^{-2k})$ rather than $\Theta(N^{-4k})$, succeeds; every classical reference in this thesis is therefore that decomposition, and the consequence of the small relative gap is the degree, not an unresolvable ground state. The second concerns a single query: one post-selected application of a block encoding of R to a Haar-random state of dimension D succeeds with mean probability $\|R\|_F^2/(D\alpha_R^2)$, which is $\Theta(1)$ for the descriptor and $\Theta(N^{-1})$ per axis for the structured standard form, whose dense $\mathbb{G}^k$ dominates the Frobenius norm. This compares the encodings only; it is not a cost of the preparation, which post-selects no single query.

5.10 Summary

The residual can be reformulated so that its Hamiltonian is banded without changing its solution, and the reformulation has a price.

1. Keeping the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ of $\mathbb{G}$ and carrying the derivatives as unknowns, as a chain for equations of order two and above or in mass form for first-order equations and systems, makes the residual banded with its kernel exactly the standard form's (Theorem 1).

2. Its block encoding is one flat LCU with $\lceil \log_2(n+4) \rceil + 8$ ancilla for R2's system rows, of the least order among exact LCU encodings, and its subnormalisation tends to the norm on regular one-axis problems (Proposition 3). Its price is a larger state, by the chain's block register, and, on the constant-coefficient problems, more gates per query at the sizes reached.

3. The Hamiltonian's norm is $\Theta(N^2)$, and for a one-axis equation whose leading coefficient does not vanish the filter's degree is $\tilde{\Theta}(N)$, against the standard form's $\tilde{\Theta}(N^{2k})$ (Equation (5.17)). Neither is polylogarithmic.

4. *The limit.* The carried derivatives erode the gap where the leading coefficient vanishes and on two axes. An a priori block rescaling restores it on one axis at no gate cost, at the price of a larger norm where the leading coefficient vanishes; on two axes the gap still falls, and a double zero is beyond any block rescaling.

§§ 9.3 and 9.4 test the cost and the gap on the benchmark problems. In one sentence: carrying the intermediate derivatives instead of eliminating them turns the physics-informed residual from dense to banded with its kernel unchanged, so that the Hamiltonian's norm falls from $\Theta(N^{4k})$ to $\Theta(N^2)$; the price is a block register and a gap that the derivative blocks erode, which an a priori rescaling restores on one axis at no gate cost.

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6 Coupled Systems and Complex-Valued Fields

6.1 Several Fields on One State

6.2 Data for Several Fields: Ratio Rows

6.3 The Mass Form for Systems

6.4 The Coupling on a Circuit

6.5 Complex-Valued Fields

6.6 From Systems to Lifts

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7 Polynomial Nonlinearity

7.1 Two Treatments

7.2 The Doubled Space: Exact Encoding, Degenerate Kernel

The doubled Hilbert space method can create degenerate kernels. QITE can't select the physical solution from amongst a set of degenerate ground states, so although it is possible to encode the solution to the original problem in the ground state of the Hamiltonian, it is not possible to extract it from the ground state of the descriptor Hamiltonian.

Proposition 4 (Imaginary time projects onto a degenerate kernel)**.**

$$\lim_{\beta\to\infty} \frac{e^{-\beta H_\otimes}\,\varphi_0}{\left\| e^{-\beta H_\otimes}\,\varphi_0 \right\|} = \frac{\Pi_{\ker H_\otimes}\,\varphi_0}{\left\| \Pi_{\ker H_\otimes}\,\varphi_0 \right\|}, \qquad \Psi = \psi \otimes \psi. \tag{7.1}$$

7.3 The Carleman Lift in Space–Time

$$\frac{du}{dt} = \sum_{p=1}^{P} F_p\, u^{\otimes p} \quad \longrightarrow \quad \frac{dz}{dt} = \mathbb{A}_K\, z, \qquad z = \big(u,\ u^{\otimes 2},\ \ldots,\ u^{\otimes K}\big), \tag{7.2}$$

$$\rho = \frac{\|u_0\|\,\|F_2\|}{\left|\operatorname{Re}\lambda_1(F_1)\right|} < 1, \qquad \text{truncation error} = O\big(\rho^{K}\big), \tag{7.3}$$

$$\begin{aligned} A_{\mathrm{st}}^{\mathrm{std}} &= \mathbb{G}_t \otimes I - \tfrac{T}{2}\, I \otimes \mathbb{A}_K, \\ A_{\mathrm{st}}^{\mathrm{desc}} &= \tilde{\mathbb{D}}_t \otimes I - \tfrac{T}{2}\, \tilde{\mathbb{B}}_t \otimes \mathbb{A}_K = \big(\tilde{\mathbb{B}}_t \otimes I\big)\, A_{\mathrm{st}}^{\mathrm{std}}, \end{aligned} \qquad \xi = \frac{2t}{T} - 1. \tag{7.4}$$

7.4 Products and Couplings on a Circuit

Proposition 5 (The nodal fold is optimal)**.**

$$\mathbb{n}_1^{(p)} = 2^{p\ell/2}\, \mathbb{U}^{\dagger}\, \Delta_p\, \mathbb{U}^{\otimes p}, \qquad \alpha\big(\mathbb{n}_1^{(2)}\big) = 2 = \big\|\mathbb{n}_1^{(2)}\big\|. \tag{7.5}$$

7.5 Fourier–Galerkin Fluids: Burgers and Navier–Stokes

$$u_t + u\,u_x = \nu\, u_{xx}, \qquad F_1 = \operatorname{diag}\big(-\nu k^2\big), \tag{7.6}$$

$$\omega_t + J(\psi,\omega) = \nu\,\nabla^2\omega, \qquad \nabla^2\psi = -\omega \ \text{ on } [0,2\pi)^2, \qquad \omega_0^{\mathrm{TG}} = 2\sin x\,\sin y. \tag{7.7}$$

7.6 Preparation Versus Projection

Table 5: Which nonlinear route prepares the solution, and which only projects onto a kernel that contains it.

Route	Kernel	What imaginary time returns	Precondition
Doubled space (Wu et al.; R6)	degenerate, dimension $\geq 2^{2n-1}$	the projection of φ_0 onto the kernel	none
Carleman lift (R8–R10), either form	one-dimensional near-kernel	the truncated solution	$\rho < 1$

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8 Preparing and Verifying the Ground State

Pipeline schematic (TikZ). Problem specification (equation, constraints, variant, n) → residual assembly (standard form | descriptor form) → stacked-residual block encoding U_R → reflection V → walk W → GQSP imaginary-time filter → post-selection → readout. Alongside: the classical reference (SVD of R) and the three verification tiers (circuit simulation, exact emulation, resource estimate) attached to the stages they check.

Figure 4: Model configuration: from problem specification to decoded field.

8.1 Problems and Model Inputs

8.2 The Classical Reference

$$|\psi_{\text{ref}}\rangle = \arg\min_{\|\psi\|=1} \|R\psi\|_2, \qquad \kappa\big(R^\dagger R\big) = \kappa(R)^2. \tag{8.1}$$

8.3 From Stacked Residual to Quantum Walk

$$R = \begin{pmatrix} R_1 \\ \vdots \\ R_L \end{pmatrix}, \qquad \big(\langle j| \otimes \langle 0_a|\big)\, U_R\, \big(|0\rangle \otimes |0_a\rangle\big) = \frac{R_j}{\alpha_R}, \qquad \alpha_R = \Big(\sum_{j=1}^{L} \alpha_j^2\Big)^{1/2}, \qquad H = R^\dagger R. \tag{8.2}$$

Table 6: The shared constraint layer: every constraint row, in either form, is one of four kinds.

Kind	Operator	Use
Zero (invariant)	$\mathbb{B}(x)$ on the chosen field or derivative block	$f(x_z) = 0$, $f'(x_m) = 0$
Collapse (regular)	$\mathbb{D}^{(0)}(x_s)$	constants, sources; lifting linear terms in the doubled space
Ratio (vector data)	$\mathbb{F} = y_a\big(I - \lvert a\rangle\langle a\rvert\big) - w\langle a\rvert$, $w = \sum_{c\neq a} y_c \lvert c\rangle$	multi-component initial data (Carleman); coupled fields
Datum slice	$\big(\mathbb{S}_a(x^*) - \hat{g}\,\tau(x_s)^\top/v\big)\,\Pi_f$	inhomogeneous boundary or initial functions, without doubling

Table 7: The benchmark ladder. Each problem adds one capability to those below it; R2–R6 are the paper's Figs 3–7.

	Problem	Reference	New capability
R1	$f' + 2f = 0$, $f(-1) = 1$	$e^{-2(x+1)}$	smallest end-to-end system
R2	Fig. 3: $\{1, 4, 4\}$, $\{1, -2, -3\}$, $\{1, 5, 400\}$	closed form	order 2; Dirichlet and Neumann invariants; stiffness
R3	Fig. 4: Legendre, $m = 0$ ($l = 4, 5$) and $m = 1$ ($l = 5, 6$)	P_l, P_l^1	variable coefficients and lifts; an exact kernel ($m = 0$)
R4	Fig. 5: four inhomogeneous equations	closed form	source injection via $\mathbb{D}^{(0)}$; Maclaurin sources
R5	Fig. 6: Laplace (2-D), heat and wave (1+1-D)	closed form	multi-axis tensors; invariants versus datum slices
R6	Fig. 7: $4f'' + 2f'^2 + f = 0$; $f'' - 2f^2 + x = 0$	$1 - x^2/8$; boundary-value solve	doubled space (kernel degeneracy)
R7	$u' = v$, $v' = -\omega^2 u$, $\omega = \pi$	sin, cos	two fields; ratio rows
R8	$y' = -y + y^2$, $y(0) = \frac{1}{2}$	$1/(1 + e^t)$	Carleman lift, both forms in time
R9	viscous Burgers, periodic, Fourier–Galerkin	Cole–Hopf; DNS	field-scale Carleman; convolution F_2; QFT fold
R10	2-D vorticity Navier–Stokes, periodic	Taylor–Green; DNS	the culmination

Proposition 6 (Stacked-residual reflection)**.**

$$\begin{aligned} V &= U_R^\dagger \big(2\Pi_{a=0} - I\big)\, U_R, \qquad V^\dagger = V, \quad V^2 = I, \\ \Pi_{\text{in}}\, V\, \Pi_{\text{in}} &= \frac{2H}{\alpha_R^2} - I \equiv H', \qquad C(V) = U_R^\dagger\, C\big(2\Pi_{a=0} - I\big)\, U_R. \end{aligned} \tag{8.3}$$

$$W = \big(2\Pi_{\text{in}} - I\big)\, V, \qquad \Pi_{\text{in}}\, W^k\, \Pi_{\text{in}} = T_k(H'). \tag{8.4}$$

8.4 The Filter and Its Parameters

$$\Pi_{\text{in}}\, P(W)\, \Pi_{\text{in}} = \sum_{k=0}^{d} q_k\, T_k(H'), \qquad P(z) = \sum_{k=0}^{d} q_k z^k, \qquad \sup_{|z|=1} |P(z)| \le 1. \tag{8.5}$$

$$f_\beta(x) = e^{-\beta(1+x)}, \qquad f_\beta(H') = e^{-2\beta H/\alpha_R^2}, \qquad q_k = \big(2 - \delta_{k0}\big)(-1)^k e^{-\beta} I_k(\beta). \tag{8.6}$$

Proposition 7 (Normalisation of the imaginary-time filter)**.**

$$\sum_{k=0}^{\infty} |q_k| = 1. \tag{8.7}$$

$$\beta = \frac{\alpha_R^2}{2\Delta}\, \ln\frac{1}{\varepsilon_f}, \qquad d \approx \sqrt{2\beta\, \ln(2/\varepsilon_t)}, \tag{8.8}$$

$$|\varphi_0\rangle \propto \bigotimes_{j=0}^{n-1} \Big(|0\rangle + r^{2^j}\, |1\rangle \Big), \qquad \gamma = \big|\langle \psi_{\text{ref}} | \varphi_0 \rangle\big|. \tag{8.9}$$

$$p_G = \big\| f_\beta(H')\, \varphi_0 \big\|^2 \;\to\; \gamma^2, \qquad \text{gates} \approx \frac{\pi}{4\gamma} \Big[2d\, g(U_R) + d\, g(\text{refl}) \Big]. \tag{8.10}$$

Optional quantikz circuit: one GQSP ancilla; $R_0 \prod_{k=1}^{d} \big[C_0 W\, R_k\big]$ with $C_0 W$ the controlled walk built from U_R, $U_R^\dagger$ and the two reflections, only the reflections controlled (Proposition 6); post-selection on the ancilla and the block-encoding register. The R_k phases realise whichever filter is in force (minimax, primary; imaginary time, the paper's).

Figure 5: The GQSP filter circuit.

8.5 Verification and the Error Budget

Table 8: Operator-verification modes: what each compares, up to what size, and to what tolerance.

Mode	Comparison	Size	Tolerance
Exact	statevector per input column; $\alpha\,\Pi_{\text{out}} U \Pi_{\text{in}}$ against the independently assembled operator	$\lesssim 20$ qubits	max-abs $\leq$ $10^{-10}\|X\|$
Probe	16–32 Haar-random and structured inputs against $X\psi/\alpha$ by sparse matrix–vector products	$\lesssim$ 28–30 qubits	relative $\leq 10^{-9}$
IR	compiled term IR against the residual assembly at every size; each term type exactly at small n	any n	$\leq 10^{-12}$
Approximation	published $\mathbb{G}$ only: entry error $\epsilon_{\mathbb{G}}(n)$ against the small-angle bound	exact sizes	reported, not gated

Table 9: Acceptance criteria for every state-preparation run.

Level	Metric	Acceptance
State	infidelity $1 - \|\langle\psi_{\text{ref}}\|\psi\rangle\|^2$ in the form's own space	$\leq \varepsilon = 10^{-6}$, and $\leq 10\times$ the predicted bound
Probability	measured p_G against $\|P(H')\varphi_0\|^2$	relative difference $\leq 10^{-8}$
Field	decoded f_Q against the analytic solution (L^∞, relative L^2)	$\leq \varepsilon_{\text{disc}} + \kappa_{\text{read}}\sqrt{\varepsilon_{\text{prep}}}$
T1 against T2	state difference	$\leq 10^{-10}$
Phases	realised against target polynomial	$\leq 10^{-12}$

$$\|f_Q - f\| \leq \underbrace{\|f_{\text{ref}} - f\|}_{\varepsilon_{\text{disc}}} + \kappa_{\text{read}}\big(\varepsilon_{\text{filter}} + \varepsilon_{\text{trunc}} + \varepsilon_{\text{phase}} + d\,\varepsilon_{\text{BE}}\big) + \varepsilon_{\text{sample}}. \tag{8.11}$$

8.6 Reading Out the Solution

$$f^*(x) = 4P_C(x) - P_G(x) - \frac{1}{2N} = \sqrt{\frac{p_G}{N}}\,\tau(x)\cdot\psi, \tag{8.12}$$

$$f_Q(x) = f(x_s)\,\frac{f^*(x)}{f^*(x_s)}, \qquad \eta_e = \frac{f(x_s)^2\,p_G}{N\,f^*(x_s)^2}. \tag{8.13}$$

8.7 Design of the Numerical Experiment

8.8 Software and Computation

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9 Numerical Study

9.1 Reproducing Wu et al.

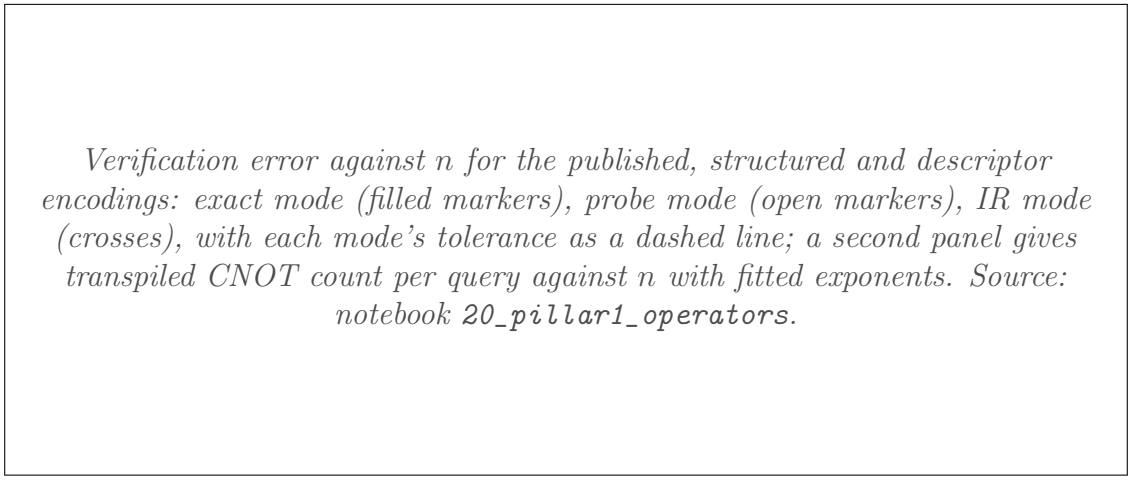
The closed form of the ideal scale factor is Equation (4.1), derived in § 4.1.

Table 10: The paper's printed scale factors against the exact solution's $\|b\|^2$ (Equation (4.1)) and the paper-faithful ground state at the printed n. †: the ground state is not trusted (a degenerate kernel, or a gap ratio below 100). Classical computation.

Fig.	Problem	n	Printed η_e	Ideal $\|b\|^2$	Ground state	Agreement
3b	R2a	3	1.29	1.29009	1.28731	✓
3c	R2b	3	32.47	32.47741	32.22636†	✓(one unit low)
3d	R2c	5	451.71	451.70962	451.52312	✓
4a	R3a	3	1.75	1.74512	1.74512	✓
4b	R3b	3	1.49	1.48779	1.48779	✓
4c	R3c	5	108.51	108.51142	109.70433†	✓
4d	R3d	5	135.38	135.38360	137.46159†	✓
5a	R4a	3	1.35	1.34649	1.34478	✓
5b	R4b	3	14.74	14.74797	14.72531	✓(one unit low)
5c	R4c	4	0.82	0.82116	0.82521	✓
5d	R4d	4	77.51	77.51167	76.93737	✓
6b	R5a	3	5.21559	5.21653	5.24407	×
6d	R5b	4	603.863	598.86606	598.87966	×
6f	R5c	5	216.469	239.82712	239.82712	×
7b	R6a	2	3.75367	3.52344	3.52344†	×
7c	R6b	2	0.07984	0.04383	0.04182†	×

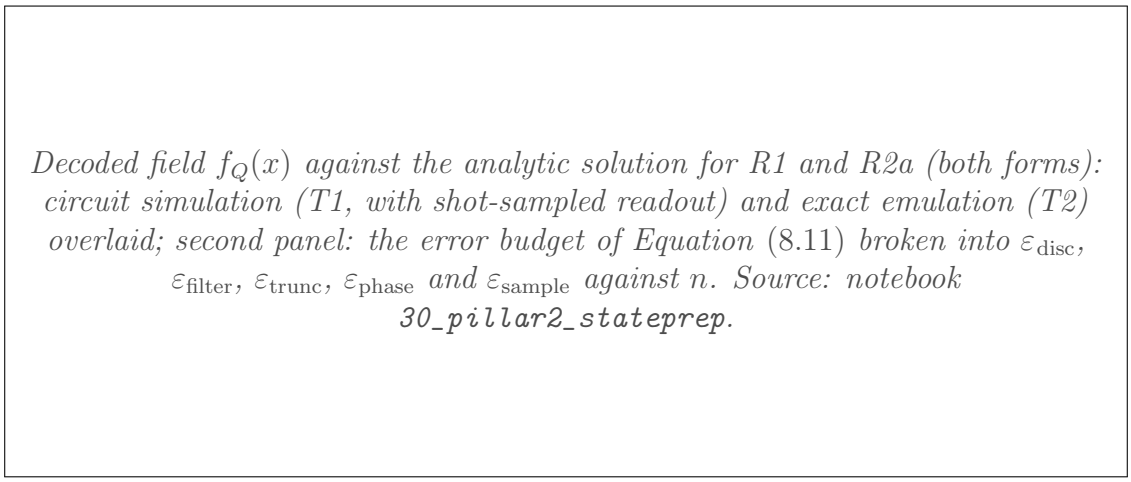
Four panels, the paper's Figs 5a–5d (R4a–R4d): the analytic solution, the paper-faithful ground state (rounded constraint points, Maclaurin sources) and R4a–R4d solved with exact roots and Chebyshev sources, at the printed n; inset or second row: L^∞ error against n for both variants. Source: notebook `10_paper_reproduction`. Classical computation.

Figure 6: Printed choices distort the paper's solutions at the printed resolution.



Verification error against n for the published, structured and descriptor encodings: exact mode (filled markers), probe mode (open markers), IR mode (crosses), with each mode's tolerance as a dashed line; a second panel gives transpiled CNOT count per query against n with fitted exponents. Source: notebook `20_pillar1_operators`.

Figure 7: Every encoding is verified to its mode's tolerance at every size tested.



Decoded field $f_Q(x)$ against the analytic solution for R1 and R2a (both forms): circuit simulation (T1, with shot-sampled readout) and exact emulation (T2) overlaid; second panel: the error budget of Equation (8.11) *broken into $\varepsilon_{\text{disc}}$, $\varepsilon_{\text{filter}}$, $\varepsilon_{\text{trunc}}$, $\varepsilon_{\text{phase}}$ and $\varepsilon_{\text{sample}}$ against n. Source: notebook* `30_pillar2_stateprep`.

Figure 8: The prepared state reproduces the analytic solution to within the predicted error budget.

Table 11: Where the cost lies: the paper's regime under its published encoding and under the structured exact encoding, against the descriptor form, for Fig. 3b ($\{1, 4, 4\}$, $f(-1) = 0$). Tier: T2 where measured, T3 beyond.

Quantity	Standard, published	Standard, structured	Descriptor
$\|H\|$	$\Theta(N^8)$	$\Theta(N^8)$	$\Theta(N^2)$
Gap Δ	7.04 (flat)	7.04 (flat)	0.454 (flat)
$\alpha_{\mathbb{G}}/\|\mathbb{G}\|_2$	$\sim N^3/2$	≤ 2.12	no $\mathbb{G}$
$\alpha_H/\|H\|$	$\sim N^{12}$	$22 \to 60$	$5.3 \to 1.05$
Gates per query	$O(n^3)$, approximate	$O(n^2)$, exact	$O(n^2)$, exact
Ancilla	$2n + 3$ ($\mathbb{G}$ alone)	$n + O(\log n)$ per $\mathbb{G}$	$\lceil \log_2(n+4) \rceil + 8$
Degree at $n = 8$ ($\varepsilon = 10^{-6}$)	8.2×10^{23}	1.2×10^{10}	5.6×10^3
Degree scaling	$\sim N^{10}$	$\tilde{\Theta}(N^4)$	$\tilde{\Theta}(N)$

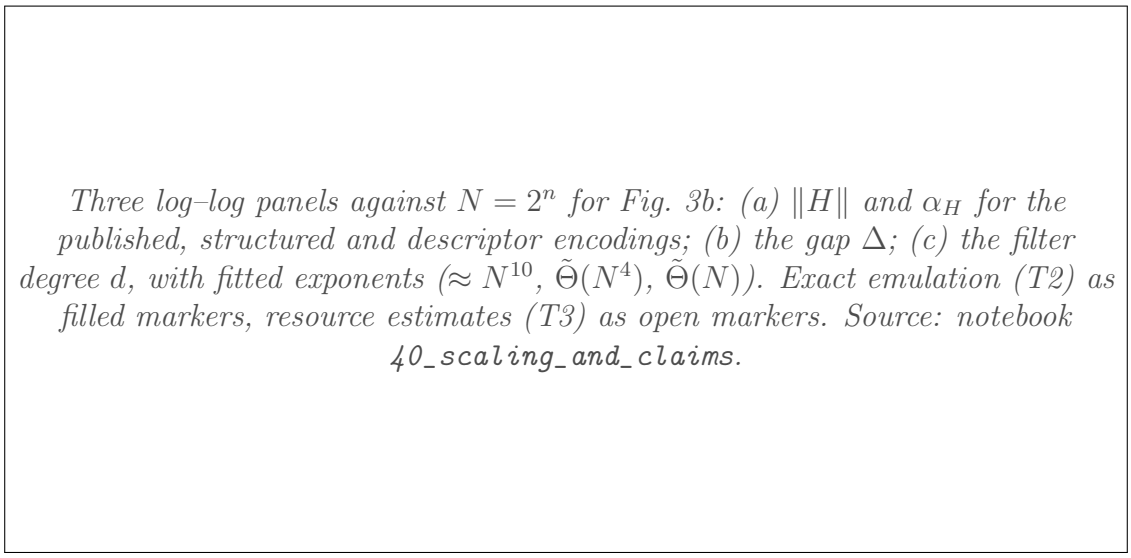


Figure 9: The encoder wall and the Hamiltonian wall are separate gaps.

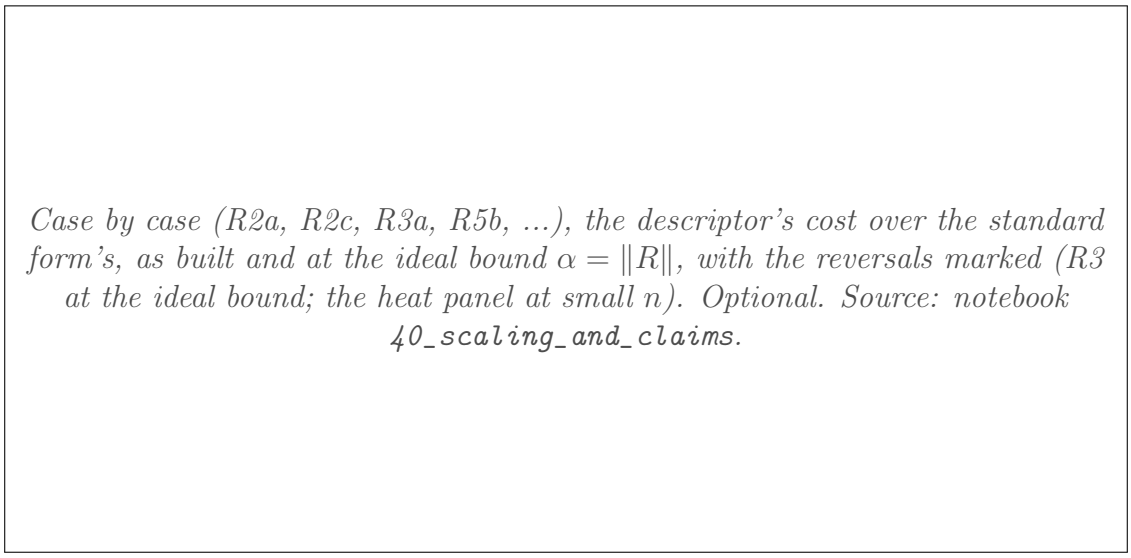


Figure 10: Three representative problems: where the comparison, quoted as built, reverses at the ideal encoding bound.

9.2 The Pipeline Is Faithful

9.3 Where the Cost Lies

9.4 The Descriptor's Gap and Its Rescaling

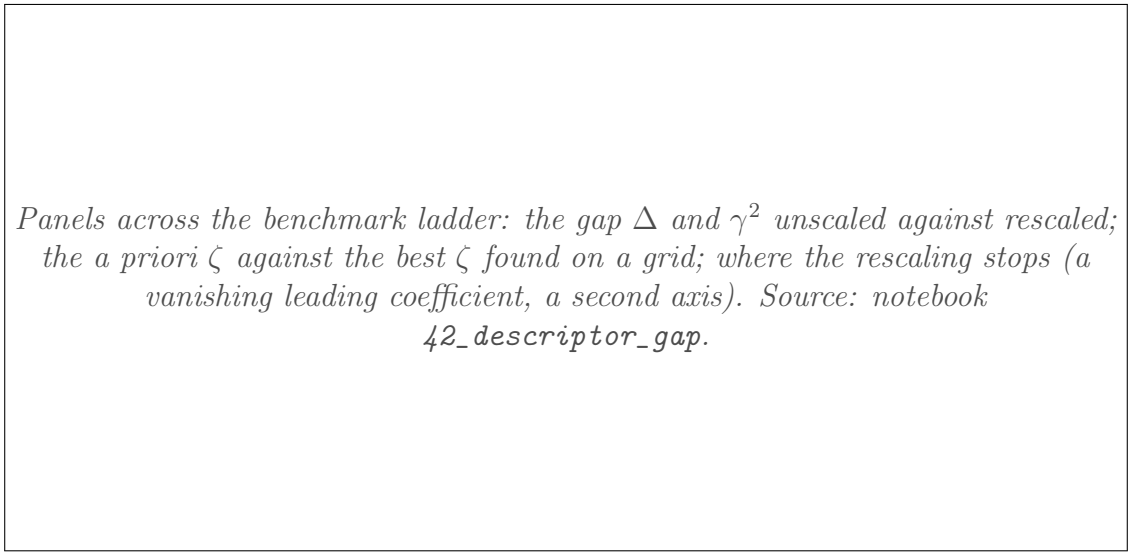


Figure 11: Block rescaling restores an O(1) gap on one axis, and stops at a double zero or on two axes.

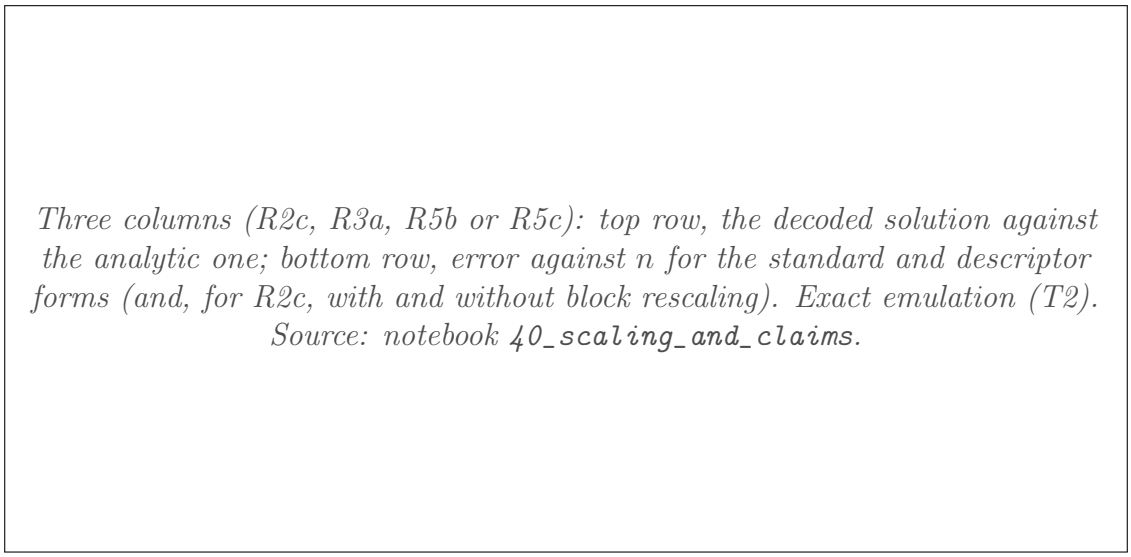
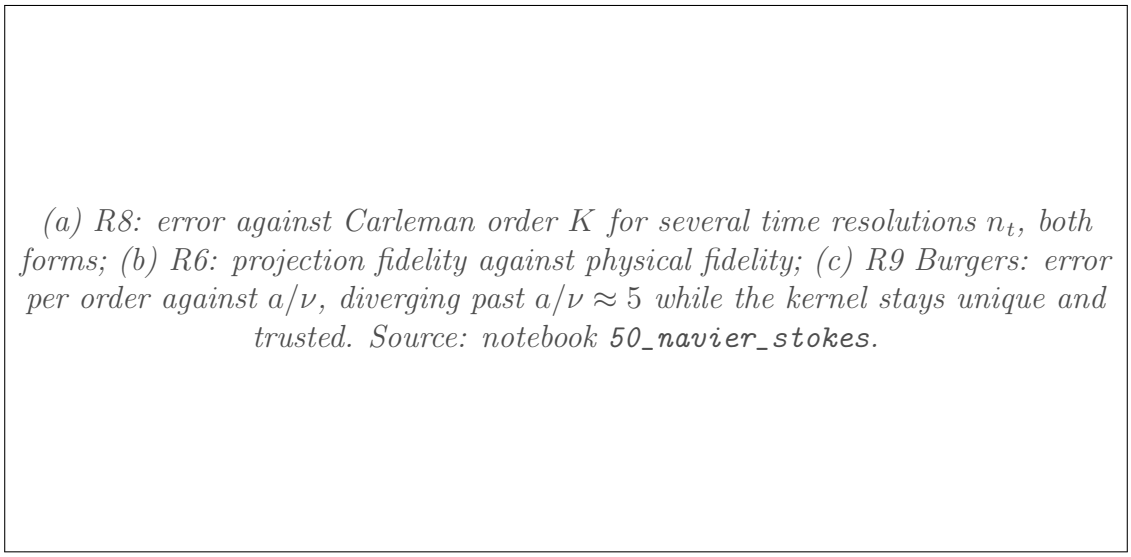


Figure 12: Three representative problems, solved in both forms.

9.5 Systems and Nonlinear Problems

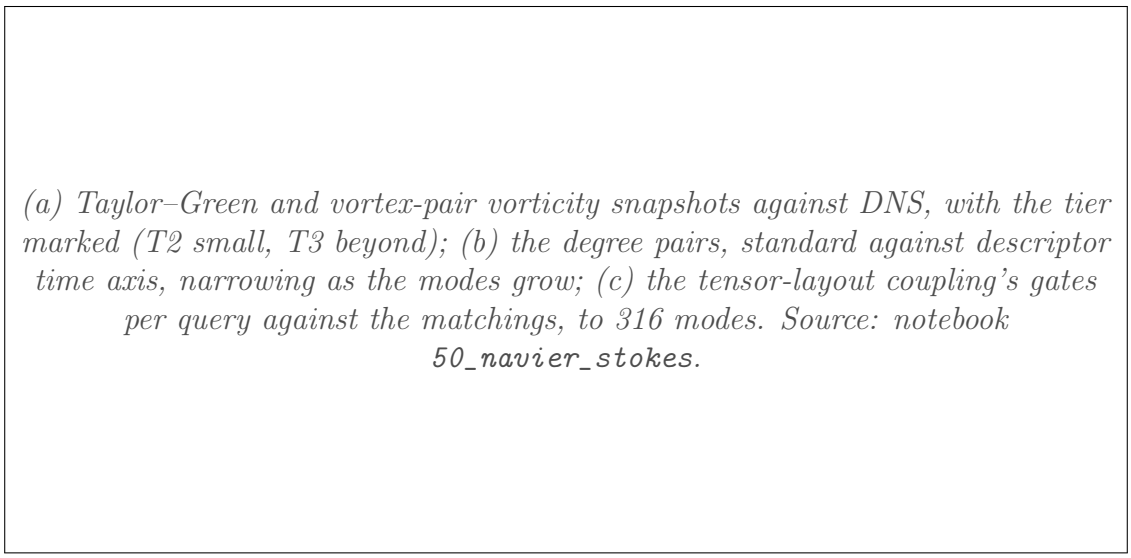
9.6 Case Study: Two-Dimensional Navier–Stokes

9.7 Summary of Findings



(a) R8: error against Carleman order K for several time resolutions n_t, both forms; (b) R6: projection fidelity against physical fidelity; (c) R9 Burgers: error per order against a/ν, diverging past $a/\nu \approx 5$ while the kernel stays unique and trusted. Source: notebook `50_navier_stokes`*.*

Figure 13: Carleman truncation converges geometrically once time is resolved; the doubled space only projects; on Burgers the error tracks a/ν, not ρ.



(a) Taylor–Green and vortex-pair vorticity snapshots against DNS, with the tier marked (T2 small, T3 beyond); (b) the degree pairs, standard against descriptor time axis, narrowing as the modes grow; (c) the tensor-layout coupling's gates per query against the matchings, to 316 modes. Source: notebook `50_navier_stokes`*.*

Figure 14: Navier–Stokes is reached by exact emulation and a resource estimate, not by circuit simulation.

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10 Discussion

10.1 Why the Descriptor Form Wins, and What It Costs

10.2 End-to-End Complexity

10.3 What the Reproduction Says About the Published Method

10.4 Where the Construction Applies

10.5 Feasibility on Fault-Tolerant Hardware

10.6 Limitations

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11 Conclusions and Future Work

11.1 Conclusions

11.2 Future Work

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12 References

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A Quantum Gate Reference

Table 12: The gates used in this thesis.

Gate	Circuit symbol	Matrix
Hadamard		
Pauli X, Y, Z		
Rotation $R_Y(\theta)$		
Phase $P(\phi)$		
CNOT		
Toffoli		
Multi-controlled X		
Zero-state reflection $2\left\|0\right\rangle\left\langle 0\right\| - I$		
Quantum Fourier transform		

B Chebyshev and Ultraspherical Identities

This appendix records the entries of the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ introduced in § 5.2, the exact factorisation $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell\geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell\geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1}\, p_{\ell-1} - p_{\ell+1} = 2\ell\, a_\ell, \qquad \ell \geq 1, \tag{B.1}$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0. Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ with $\tilde{\mathbb{D}}_{\text{raw}}$ a single superdiagonal, $(\tilde{\mathbb{D}}_{\text{raw}})_{\ell,\ell+1} = 2(\ell+1)$, and $\tilde{\mathbb{B}}_{\text{raw}}$ two diagonals, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell} = c_\ell$, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell+2} = -1$.

B.2 The normalised basis

The construction of § 5.2 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ $(\ell \geq 1)$, orthonormal under the discrete N-point Chebyshev inner product $\langle T_k, T_\ell\rangle_N = \frac{N}{2}\, c_\ell\, \delta_{k\ell}$ [52]: since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \operatorname{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{\mathbb{B}} = S\,\tilde{\mathbb{B}}_{\text{raw}}\,S^{-1}$ and $\tilde{\mathbb{D}} = S\,\tilde{\mathbb{D}}_{\text{raw}}\,S^{-1}$, that is

$$\tilde{\mathbb{D}}_{\ell,\ell+1} = 2(\ell+1)\sqrt{c_\ell/c_{\ell+1}}, \qquad \tilde{\mathbb{B}}_{\ell,\ell} = c_\ell, \qquad \tilde{\mathbb{B}}_{\ell,\ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \tag{B.2}$$

so $\tilde{\mathbb{D}}$ has the single non-zero diagonal $(2\sqrt{2},\ 4,\ 6,\ 8,\ \dots)$ and $\tilde{\mathbb{B}}$ has main diagonal $(2,\ 1,\ 1,\ \dots)$ and second superdiagonal $(-\sqrt{2},\ -1,\ -1,\ \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries at any N.

The dense Chebyshev derivative matrix is their exact quotient,

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{B.3}$$

an algebraic identity rather than a numerical approximation: $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ are read directly off the same recurrence that defines $\mathbb{G}$ (Equation (B.1)), so the equality holds exactly at every N, independent of whichever values a test suite happens to check. Appendix I reports this identity's residual against the reference implementation, alongside the rest of the composition-algebra checks.

B.3 The norm of the derivative matrix

B.4 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [4]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1-x^2)^{\lambda-1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{\mathrm{d}T_k}{\mathrm{d}x} = k\, C_{k-1}^{(1)}, \qquad k \geq 1 \tag{B.4}$$

[4, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u''s coefficients now live in a different basis to u's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ $(k=0)$, $\frac{1}{2}$ $(k \geq 1)$, and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [4, Eq. 2.8]. Comparing entries, $\tilde{\mathbb{D}}_{\text{raw}} = 2\mathcal{D}_0$ and $\tilde{\mathbb{B}}_{\text{raw}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ of Equation (B.1) is literally $\mathcal{S}_0\, p = \mathcal{D}_0\, a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of § 5.2 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [4] raises the basis one level per derivative: a λ-th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $\mathrm{d}C_k^{(\lambda)}/\mathrm{d}x = 2\lambda\, C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \to C^{(\lambda+1)}$, still two diagonals [4, Eqs. 3.2–3.3]. To assemble one equation of order k, every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1}\cdots\mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $\mathrm{L} = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1}\cdots\mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved

once for the coefficients of $u^{(k)}$ [4, §3.2] – so a k-th order equation needs k distinct conversion operators $\mathcal{S}_0, \ldots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of § 5.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $\hat{f}^{(j)}$ of w, and reuses the single $\lambda = 0$ pair $(\tilde{\mathbb{B}}, \tilde{\mathbb{D}})$ between every consecutive pair of blocks (Equation (5.6)). No elimination is performed, and no operator beyond $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$ and the multiplication operators of Appendix G.2 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix G.2). What this thesis adds to [4] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1], whose block structure is exactly what § 5.7 block-encodes.

B.5 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{\mathbb{B}} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \tilde{\mathbb{D}} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbb{G} = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad \text{(B.5)}$$

The top-right entry $3\sqrt{2}$ of $\mathbb{G}$ is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of $\mathbb{G}$ is non-zero at columns $\ell + 1, \ell + 3, \ldots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{\mathbb{B}}$ or $\tilde{\mathbb{D}}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ of § 5.3, $w = (\hat{f}^{(0)}; \hat{f}^{(1)})$ and $A_{\text{desc}} = \left(\begin{smallmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ 2I_4 & I_4 \end{smallmatrix}\right)$ is the 8×8 matrix

$$A_{\text{desc}} = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right]. \quad \text{(B.6)}$$

The top four rows are the recurrence block $-\tilde{\mathbb{D}}\,\hat{f}^{(0)} + \tilde{\mathbb{B}}\,\hat{f}^{(1)} = 0$; the bottom four are the differential row $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$. Every non-zero entry lies on a diagonal or a near-diagonal, in place of the dense $\mathbb{G} + 2I$ of the eliminated form.

C The Standard Form of Wu et al. in Full

This appendix gives the $N = 4$ instance of the standard-form residual A_{std} that § 3.2 builds for Wu et al.'s constant-coefficient case, and the entries of the lifting operator $\mathbb{M}_{x^p}$ that the same section uses for a variable coefficient.

Why a condition for all x is a finite linear system. A condition required at every $x \in [-1, 1]$, such as the differential equation, reaches the form $\langle\tau(x)|\, C\,|\psi\rangle = 0$ for all x. The components of $\langle\tau(x)|$ are linearly independent functions of x, so $\langle\tau(x)|\,\mathbf{v}$ vanishes for all x only if $\mathbf{v} = 0$. Taking $\mathbf{v} = C\,|\psi\rangle$, the condition for all x is therefore equivalent to the finite system $C\,|\psi\rangle = 0$, which no longer depends on x. This is the step that § 3.2 states without proof.

The standard-form residual at $N = 4$. For Wu et al.'s stiff pilot equation [1], $f'' + 5f' + 400f = 0$, Equation (3.3) gives $A_{\text{std}} = \mathbb{G}^2 + 5\mathbb{G} + 400I$. At $N = 4$, using $\mathbb{G}$ (Appendix B),

$$A_{\text{std}} = \begin{pmatrix} 400 & 5\sqrt{2} & 4\sqrt{2} & 15\sqrt{2} \\ 0 & 400 & 20 & 24 \\ 0 & 0 & 400 & 30 \\ 0 & 0 & 0 & 400 \end{pmatrix}, \tag{C.1}$$

verified directly against `pihm.chebyshev_derivative_matrix`. A_{std} is triangular here only because this equation has no odd-order term beyond f' at even N; the off-diagonal entries sit three orders of magnitude below the diagonal, a large-coefficient conditioning problem stacked on top of $\mathbb{G}$'s own density.

The lifting operator $\mathbb{M}_{x^p}$. $\mathbb{M}_{x^p}$ is exact and *rectangular*: multiplying a degree-$(N-1)$ model by x^p can raise its degree by as much as p, so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$\mathbb{M}_{x^1} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}, \qquad \mathbb{M}_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \tag{C.2}$$

verified directly against `pihm.multiplication_matrix`. The general entry follows the same product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ that § 3.2 states in closed form.

C.1 The faithful rebuild's conventions

C.2 The standard form on several axes

Stub: write this sub-sec-

§ 3.2 writes the two-variable case. For ω variables $\mathbf{x} = (x_1, \ldots, x_\omega)$, each carried on its own

n-qubit Chebyshev axis u, the state lives on ωn qubits and the feature vector is $|\tau(\mathbf{x})\rangle = |\tau(x_1)\rangle_n \otimes \cdots \otimes |\tau(x_\omega)\rangle_n$, so that $f_q(\mathbf{x}) = \langle\tau(\mathbf{x})|\psi\rangle$. The derivative along axis u is $\mathbb{G}_u = I^{\otimes(u-1)} \otimes \mathbb{G} \otimes I^{\otimes(\omega-u)}$; these act on different factors and commute, and a derivative of multi-index $\boldsymbol{\chi} = (\chi_1, \ldots, \chi_\omega)$ is $\partial^{\boldsymbol{\chi}} \mapsto \mathbb{G}^{\boldsymbol{\chi}} = \prod_u \mathbb{G}_u^{\chi_u} = \bigotimes_u \mathbb{G}^{\chi_u}$.

Each term t of the equation has a constant c_t, a multi-index $\boldsymbol{\chi}_t$ and a coefficient m_t that is a polynomial in a single variable x_{u_t} (the class the code builds; a coefficient that depends on several variables is not built). An axis is *lifted* when it carries a non-constant coefficient or when the equation has a source, in which case every axis is lifted. The term acts on axis u as $\mathbb{M}_{m_t}\mathbb{G}^{\chi_{t,u}}$ if $u = u_t$, as $\mathbb{M}_1\mathbb{G}^{\chi_{t,u}}$ on a lifted axis without a coefficient, and as $\mathbb{G}^{\chi_{t,u}}$ on an unlifted axis, so that

$$A_{\text{std}} = \sum_t c_t \bigotimes_{u=1}^{\omega} \mathbb{M}_t^{(u)}\, \mathbb{G}^{\chi_{t,u}} \;-\; \mathbb{M}_r\, \mathbb{D}^{(0)}(\mathbf{x}_s), \tag{C.3}$$

with $\mathbb{M}_t^{(u)}$ as just described. The source term is the rank-one operator $\mathbb{M}_r\mathbb{D}^{(0)}(\mathbf{x}_s) = |\hat{r}\rangle \langle\tau(\mathbf{x}_s)| / y_s$, with $|\hat{r}\rangle$ the coefficients of r on the lifted register (the Kronecker product of the one-dimensional coefficient vectors where r is a product of single-variable polynomials). The equation block's output register has $\omega n + \omega_{\text{lift}}$ qubits, ω_{lift} being the number of lifted axes. For $\omega = 1$, Equation (C.3) is Equation (3.4).

An invariant constraint $\partial_u^j f = 0$ on the face $x_u = x_z$ is the row block $I^{\otimes(u-1)} \otimes \mathbb{B}_n(x_z)\mathbb{G}^j \otimes I^{\otimes(\omega-u)}$, or $\mathbb{B}_{n+1}(x_z)\mathbb{M}_1\mathbb{G}^j$ on its own axis if that axis is lifted. The identity factors are on n qubits whether or not their axes are lifted, so only the equation block, not each constraint, acts into the full lifted register. The regular datum at a point $\mathbf{x}_s$ is the rank-one row $\mathbb{D}^{(0)}(\mathbf{x}_s) = \bigotimes_u \mathbb{B}_n(x_{s,u})/y_s = 2^{\omega n/2}\,|0\rangle \langle\tau(\mathbf{x}_s)|\,/y_s$. Each $\mathbb{G}_u$ has the norm $\Theta(N^2)$ of $\mathbb{G}$.

C.3 Datum-slice rows

C.4 The substitution for R3c and R3d

C.5 The uniqueness count for an ordinary equation

Proposition 8 (Uniqueness count)**.** *Let $\sum_{j=0}^{k} a_j f^{(j)} = 0$ on $[-1, 1]$ with smooth coefficients and a_k nowhere zero, and let $c_1, \ldots, c_m$ be invariant constraints (each sets a derivative of f at a point to zero). The solutions that satisfy all of them form a space of dimension $k - \rho$, where ρ is the rank of the c_i restricted to the k-dimensional solution space S. The ground state of the constraint energy is unique up to scale exactly when $\rho = k - 1$.*

Proof. Since a_k has no zero, the equation is a regular linear system of order k and S has dimension k. The map $f \mapsto (c_i(f))_{i \le m}$ is linear on S with rank ρ, so its kernel, the solutions that satisfy every constraint, has dimension $k - \rho$. A ground state of zero energy is a nonzero element of that kernel, unique up to scale exactly when the kernel is one-dimensional. For $\rho = k$ the kernel is $\{0\}$ and no nonzero state has zero energy. For $\rho < k - 1$ it has dimension at least two. □

The rank is that of the constraints restricted to S, not of the functionals, so $k-1$ constraints chosen without regard to the equation can fail. The regular datum fixes the scale of the selected solution and requires that it not vanish at the regular point. The proposition concerns the continuous problem: the assembled residual follows the count only as far as the truncation resolves S, and the dimension and gap of its kernel are computed, not inferred (§ 4.3). For an inhomogeneous equation, or one whose leading coefficient vanishes (R3a), no count is claimed here and uniqueness is judged by the computed kernel.

The Laplace invariant set. The modes $\sin\frac{\ell\pi(x+1)}{2}\,\sinh\frac{\ell\pi(y+1)}{2}$, $\ell\geq 1$, all satisfy the equation and its three homogeneous sides, and only $\ell=1$ matches the top-side function, which the printed set does not impose. This is the continuous kernel that the truncation resolves as n grows.

C.6 The paper's printed matrices at $n=2$

C.7 Lifts and folds

D The Published Circuits as Built

§ 3.3 describes, at the level a reader needs to follow the argument, the circuit Wu et al. use to load $\mathbb{G}$'s column-linear entries as a rotation amplitude [1]. This appendix gives the mechanism in full, together with a rigorous bound on its entrywise error that this thesis derives and verifies – the source paper states only that the error is "negligible" and decreases "exponentially" with n, without a closed form.

D.1 The circuits as built

The constraint rows (Figs 2 and 8). $\mathbb{B}(x)$ and $\mathbb{D}^{(0)}(x_s)$ are block-encoded by the inverse feature map $U_\tau^\dagger(x)$ followed by $\hat{S}_{n+1}=2\,|0\rangle\langle 0|-I$ controlled inside a Hadamard sandwich on one ancilla, a two-term LCU on $n+2$ qubits. The encoded block is $\mathbb{B}(x)/\sqrt{2^{n+1}}$, and $\mathbb{D}^{(0)}(x_s)/\big(\sqrt{2^{n+1}}/f(x_s)\big)$ for the collapse row. The global phase of -1 in $\hat{S}_{n+1}$ is a relative phase once the circuit is controlled, and cannot be dropped.

The derivative circuit (Fig. 9). The registers are, in order, a flag qubit acted on by $R_Y(\omega_c)$ with $\omega_c=2\arccos\sqrt{1/3}$, an X ancilla, an offset register of $m=n$ qubits sandwiched between Hadamards, a value ancilla, and the n-qubit system. Two conditioned subroutines assign the entries on different bases. In $\hat{U}_R$ the offset $|i\rangle$ selects the $(2i+1)$th superdiagonal and the system $|j\rangle$ the jth column; it consists of n singly-controlled $R_Y(\theta_k)$ gates on the value ancilla, controlled by the bits of $|j\rangle$, followed by n cyclic shifts $\hat{R}_n$. In $\hat{U}_C$ the offset selects the $(i+1)$th row; it consists of n multi-controlled $R_Y(\varphi_k)$ gates, one shift $\hat{L}_n$ and $2n$ swaps. With $\mu=2/\|\mathbb{G}\|_S$ the angles are $\theta_k=2\arcsin(-2^{k-1}\mu)$ and $\varphi_k=2\arcsin[-2^{k-1}\mu(1/\sqrt{2}-1)]$. The circuit uses

$m + n + 3 = 2n + 3$ qubits, its gate count is linear in n before the multi-controlled gates are decomposed and $O(n^3)$ after an $O(n^2)$ decomposition of each, and $\mathbb{G}$ is the top-left $2^n \times 2^n$ block of the unitary multiplied by $(2^{n-1} + 2^n)\|\mathbb{G}\|_S$. These are the paper's statements; their tests follow.

The claims tested. Table 13 gives the verdict on each claim the paper makes for its circuits. The gate count holds as a bound, the error holds to leading order in μ (Proposition 9), and the rows of Figs 2 and 8 hold exactly. The qubit count of Fig. 9 is not testable: the rebuild uses registers of its own, so its $2n+5$ neither confirms nor refutes the printed $2n+3$. The prefactor is valid but exceeds $\|\mathbb{G}\|_2$ by a factor growing as N^3 (§ 4.5.1).

Table 13: **The paper's encodings hold their gate and error claims but not their efficiency.** The claims of Wu et al. for their block encodings [1], tested against the rebuilt circuits. Counts are for the rebuilt circuits; the error is to leading order in μ.

Circuit and claim	Printed	As rebuilt	Verdict
Fig. 2, $\mathbb{B}(x)$	$n+2$ qubits, $\sqrt{2^{n+1}}$	the same, exact	holds
Fig. 8, $\mathbb{D}^{(0)}(x_s)$	$\sqrt{2^{n+1}}/f(x_s)$	the same, exact	holds
Fig. 9, qubits	$2n+3$	$2n+5$ (own registers)	not testable
Fig. 9, gates	$O(n^3)$	$O(n^2)$	holds (as a bound)
Fig. 9, error	decreases exponentially	$O(N^{-5})$, bound derived	holds (leading order)
Fig. 9, prefactor	$(2^{n-1} + 2^n)\|\mathbb{G}\|_S$	valid, $\sim j_{-3/4,1} N^3 \|\mathbb{G}\|_2$	loose

D.2 The small-angle ladder and its error

The mechanism. A ladder of n singly-controlled R_y rotations acts on one flag qubit, initialised to $|1\rangle$: bit m of the (register-arithmetic-shifted) column index controls $R_y(\theta_m)$,

$$\theta_m = 2\arcsin\big(-2^m\mu\big), \qquad m = 0, \ldots, n-1, \tag{D.1}$$

for a small real coefficient μ. Because rotations on the same qubit compose by exact angle addition ($R_y(a)R_y(b) = R_y(a+b)$, an $SO(2)$ identity), the net rotation for a column index p with bit set S is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m\in S}\theta_m$, giving the flag qubit's $|0\rangle$-amplitude

$$\mathrm{raw}(p) = -\sin\big(\Theta(p)/2\big) = \sin\Big(\textstyle\sum_{m\in S}\arcsin(2^m\mu)\Big). \tag{D.2}$$

The row-0 correction (entries $\sqrt{2}p$ in place of $2p$) is a second, independently accumulated ladder on the same flag qubit, with per-bit coefficient $\mu(1/\sqrt{2}-1)$, controlled on the same "this is row 0" flag the main construction already computes.

Why this is only approximate. $\mathrm{raw}(p)$ in Equation (D.2) would equal the desired linear value μp exactly only if $\sin\big(\sum_m \arcsin(x_m)\big) = \sum_m x_m$, which does not hold in general – sin of a sum is not the sum of sines. The discrepancy is a genuine, quantifiable small-angle error.

Proposition 9 (Entrywise error of the ladder construction)**.** *Let* $J_3(p) := \sum_{m\in S}(2^m)^3$ *for column index* p *with bit set* S*. Then*

$$\left|\mathrm{raw}(p) - \mu p\right| \;\le\; \frac{\mu^3}{6}\,p^3 \;+\; O(\mu^5 N^5). \tag{D.3}$$

Proof. Expand both non-linearities to leading order in the small parameter μ: $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first expansion over the fired bits and substituting into the second,

$$\sum_{m\in S}\arcsin(2^m\mu) = \mu p + \tfrac{\mu^3}{6}J_3(p) + O(\mu^5 N^5), \tag{D.4}$$

$$\begin{aligned}\sin\Big(\sum_{m\in S}\arcsin(2^m\mu)\Big) &= \sin(\mu p) + \cos(\mu p)\cdot\tfrac{\mu^3}{6}J_3(p) + O(\mu^5 N^5)\\ &= \mu p + \tfrac{\mu^3}{6}\big(J_3(p) - p^3\big) + O(\mu^5 N^5).\end{aligned} \tag{D.5}$$

Since $p = \sum_{m\in S} 2^m$ and every $2^m \le p$, the multinomial expansion of p^3 term-by-term dominates $J_3(p)$, with equality only when S is a single bit (p itself a power of two): $J_3(p) \le p^3$. Taking the worst case $p = N - 1$ gives Equation (D.3). □

This is a leading-order bound, valid once the largest single argument, $2^{n-1}\mu$, is small and well inside arcsin's domain – true from $n = 2$ onward at the default $\mu = 2/\|\mathbb{G}\|_S$ (Equation (3.6)), but not at $n = 1$, where $\mu \cdot 2^0 = \mu = 1$ exactly and the bound is not yet meaningful.

D.3 Measured resources and error

Table 14: The published encoding of $\mathbb{G}$ against the paper's claims. Classical exact-mode measurement.

n	N	$\|\mathbb{G}\|_S$	Qubits (claim $2n+3$)	Gates	Bound on $\epsilon_{\mathbb{G}}$	Measured $\epsilon_{\mathbb{G}}$
2	4	6.15×10^1	–	–	–	–
3	8	1.18×10^3	–	–	–	–
4	16	2.04×10^4	–	–	–	–
5	32	3.38×10^5	–	–	–	–
6	64	5.50×10^6	–	–	–	–
7	128	8.88×10^7	–	–	–	–
8	256	1.43×10^9	–	–	–	–

E The Structured Exact Encoding of the Differentiation Matrix

E.1 Proof of Proposition 1

E.2 The arithmetic SELECT

E.3 Proof of Proposition 2

E.4 Subnormalisation across n

Table 15: Subnormalisation of the structured and published encodings of $\mathbb{G}$, relative to $\|\mathbb{G}\|_2$.

n	$\|\mathbb{G}\|_2$	Structured	Published
2	7.39	1.623	49.9
3	30.1	1.858	468
4	121	1.986	4.05×10^3
5	484	2.051	3.36×10^4
6	1935	2.084	2.73×10^5
7	7739	2.100	2.20×10^6
8	30 957	2.109	1.77×10^7
9	123 827	2.113	1.42×10^8
10	495 308	2.115	1.14×10^9

F The Shared Pipeline: Identities and Formulas

F.1 Proof of Proposition 6

F.2 Qubitisation: Chebyshev polynomials of the walk

F.3 The controlled walk

F.4 The imaginary-time filter

$$e^{-\beta x} = I_0(\beta) + 2\sum_{k=1}^{\infty}(-1)^k I_k(\beta)\, T_k(x). \tag{F.1}$$

F.5 GQSP phases

F.6 The readout identity for probabilistic preparation

G The Descriptor Construction in Full

G.1 Block rows in full

G.2 Higher order, variable coefficients, and the padded block count

For an order-k equation the state is $w = (\hat{f}^{(0)}; \ldots; \hat{f}^{(k)})$ and A_{desc} stacks k recurrence block-rows $-\tilde{\mathbb{D}}\,\hat{f}^{(j)} + \tilde{\mathbb{B}}\,\hat{f}^{(j+1)} = 0$ above the single differential row $\sum_j \mathbb{M}_{a_j}\,\hat{f}^{(j)} = 0$: $k+1$ block-rows, each N coefficients wide, giving the classical $(k+1)N \times (k+1)N$ square matrix of Definition 11.

A variable coefficient $a_j(x)$ enters through $\mathbb{M}_{a_j}$, the square Galerkin truncation to the first N modes of the exact product (the multiplication of § 3.2), built from its Chebyshev coefficients, or from its monomial coefficients converted exactly. $\mathbb{M}_{a_j}$ is banded, with bandwidth the degree of a_j, so it folds into the same shift-and-diagonal form as $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$. The truncation is the only approximation in the linear construction; its error is confined to the top Chebyshev modes of a product that is smooth and resolved.

Padding to a power of two. The classical $(k+1)N \times (k+1)N$ matrix above is exact and complete as a linear system; it is not, however, the object any later stage of the pipeline actually builds. A gate-level realisation must index which block $\hat{f}^{(j)}$ a given operator acts on with a qubit register, and a register of m qubits addresses 2^m blocks, not $k+1$ of them for an arbitrary k. A_{desc} is therefore *embedded* in a $2^{\lceil \log_2(k+1) \rceil} N$-dimensional space: the $k+1$ block-rows and block-columns above are kept exactly as derived, and $2^{\lceil \log_2(k+1) \rceil} - (k+1)$ further diagonal blocks are appended, each a row $c_{\text{pad}} I_N$ with $c_{\text{pad}} = 1$ pinning its own dummy field to zero, $\hat{f}^{(j)}_{\text{dummy}} = 0$; no other row reads a padding block, so the padding decouples. This padding is exact: the physical solution already satisfies the dummy rows trivially, since it has nothing in those blocks, so no error is introduced beyond the multiplication's truncation above. The qubit cost, however, is not uniform in k:

Order k	Blocks $k+1$	Field qubits $\lceil \log_2(k+1) \rceil$	Padded blocks $2^{\lceil \log_2(k+1) \rceil}$	Dummy blocks
1	2	1	2	0
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

so the state w the gate-level construction actually carries has $2^{\lceil \log_2(k+1) \rceil} N$ components, not $(k+1)N$: equal to the unpadded count exactly when $k+1$ is itself a power of two (orders

$1, 3, 7, \dots$), and up to nearly double it otherwise – order 2, the most common case after order 1, already pays for a field register it only three-quarters uses. This is the qubit cost that must always accompany the gate-count comparison of § 5.8.

G.3 Proof of Theorem 1

G.4 The term IR and its compilation

G.5 Proof of Proposition 3

G.6 Block rescaling

H Systems and the Nonlinear Route in Full

H.1 Systems in Full

H.2 The doubled space

§ 3.4 states the published construction as quadratic only. A term of degree p in the unknown needs p copies, $|\Psi\rangle = |\psi\rangle^{\otimes p}$, of dimension N^p, with its p factors folded back into one Chebyshev series by $p - 1$ applications of the product rule (a cubic term by a tripled space and two folds, a quartic by a quadrupled space and three). The register therefore grows with the degree, a limitation noted in the concurrent work of Paine [7], and the argument of Proposition 4 should carry over, since the kernel of the corresponding Hamiltonian is again a subspace and the p-fold product states are not; this thesis does not use it beyond $p = 2$. The alternative it builds for higher degrees is the Carleman lift of § 7.3.

H.3 Proof of Proposition 4

H.4 The Carleman lift

H.5 Ratio rows and the space-time forms

H.6 Product folds

H.7 Fourier–Galerkin operators

I Verification Protocol

I.1 Modes, sizes and tolerances

I.2 Structured probes

I.3 Per-atom residuals

I.4 The classical reference: eigh against SVD

Table 16: Accuracy of the classical reference, $1 - |\cos|$ against the exact coefficients, for the standard form by `eigh`$(R^\top R)$ and by SVD of R, and for the descriptor form.

n	Standard, `eigh`	Standard, SVD	Descriptor
2	1.7×10^{-1}	1.7×10^{-1}	1.6×10^{-2}
3	8.6×10^{-8}	8.6×10^{-8}	5.8×10^{-9}
4	0	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
5	3.7×10^{-15}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
6	1.7×10^{-10}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
7	1.2×10^{-4}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
8	0.73	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$

Table 17: The linear benchmark problems, R1–R5, with the paper's printed invariant constraints.

	Equation	Invariant (printed)	Reference
R1	$f' + 2f = 0$	– ($f(-1) = 1$ regular)	$e^{-2(x+1)}$
R2a	$f'' + 4f' + 4f = 0$	$f(-1) = 0$	$\frac{1}{2}(1+x)e^{-2x}$
R2b	$f'' - 2f' - 3f = 0$	$f(0) = 0$	$\frac{1}{4}\left(e^{3x} - e^{-x}\right)$
R2c	$f'' + 5f' + 400f = 0$	$f'(0) = 0$	$e^{-5x/2}\left(\cos\omega x + \frac{\sqrt{7}}{21}\sin\omega x\right)$, $\omega = \frac{15\sqrt{7}}{2}$
R3a	Legendre, $l = 4$	$f'(0) = 0$	P_4
R3b	Legendre, $l = 5$	$f(0) = 0$	P_5
R3c	associated Legendre, $l = 5$, $m = 1$	$f'(0) = 0$	P_5^1
R3d	associated Legendre, $l = 6$, $m = 1$	$f(0) = 0$	P_6^1
R4a	$(x-1)f'' - xf' + f = (x-1)^2$	$f'(-0.2) = 0$	closed form
R4b	$f'' + 4f' + 4f = e^{-2x}$, $\bar{p} = 3$	$f(-0.73) = 0$	closed form
R4c	$f'' - 5f' + 6f = xe^{2x}$, $\bar{p} = 5$	$f(0.91) = 0$	closed form
R4d	$f'' + 4f' + 20f = xe^{-2x}\sin 4x$, $\bar{p} = 7$	$f'(-0.75) = 0$	closed form
R5a	Laplace, $f_{xx} + f_{yy} = 0$	$f(\pm 1, y) = f(x, -1) = 0$	$\cos\frac{\pi x}{2}\,\sinh\frac{\pi(y+1)}{2}/\sinh\pi$
R5b	heat, $f_t = \frac{1}{25}f_{xx}$	$f(t, \pm 1) = 0$, $f_x(t, \pm\frac{3}{4}) = 0$	$e^{-4\pi^2 t/25}\sin 2\pi x$
R5c	wave, $f_{tt} = 4f_{xx}$	$f(\frac{1}{8}, x) = f(\frac{7}{8}, x) = 0$, $f(t, \pm 1) = 0$	$\cos 4\pi t\,\sin 2\pi x$

Table 18: The nonlinear and coupled benchmark problems, R6–R10.

	Equation	Data	Reference
R6a	$4f'' + 2f'^2 + f = 0$ (paper Eq. 29)	$f'(0) = 0$, $x_s = 0$	$1 - x^2/8$
R6b	$f'' - 2f^2 + x = 0$ (paper Eq. 34)	$f(0.02615) = 0$	boundary-value solve to 10^{-12}
R7	$u' = v$, $v' = -\omega^2 u$, $\omega = \pi$	$u(-1) = 0$, $v(-1) = \omega$	$u = \sin\omega(x+1)$
R8	$y' = -y + y^2$, $t \in [0, 3]$	$y(0) = \frac{1}{2}$ ($\rho = 0.5$)	$1/(1+e^t)$
R9	viscous Burgers, periodic, $u_0 = a\sin x$	Fourier–Galerkin, $\lvert k\rvert \le K_c$	Cole–Hopf; DNS; exact lift
R10a	2-D vorticity Navier–Stokes, $\omega_0 = 2\sin x\sin y$	$J \equiv 0$ (Taylor–Green)	exact viscous decay
R10b	2-D vorticity Navier–Stokes, vortex pair and resonant triads	active nonlinearity	DNS; exact lift

Table 19: State preparation by problem and form. Tier as marked.

	n	Infidelity (std)	Infidelity (desc)	Degree (std)	Degree (desc)	Tier
R1		–	–	–	–	
R2a		–	–	–	–	
R2b		–	–	–	–	
R2c		–	–	–	–	
R3		–	–	–	–	
R4		–	–	–	–	
R5		–	–	–	–	
R7		–	–	–	–	
R8		–	–	–	–	
R9		–	–	–	–	
R10		–	–	–	–	

I.5 The circuit-simulation memory and time model

I.6 Acceptance and the error budget

J Benchmark Catalogue

J.1 Linear problems

J.2 Nonlinear and coupled problems

J.3 Results by problem

K Reproduction Record

K.1 All panels

K.2 The Fig. 5 triage

K.3 Panels not reproduced, and printed errors

K.4 The paper's claims

K.5 The paper's filter and its parameters

The paper realises imaginary time by QSVT on a block encoding of $\tilde{H} = H/\|H\|_F$ [1]. The target $p(x) \approx e^{-tx}$ has indefinite parity, so the paper finds one set of phase angles for its even part and one for its odd part, of maximal degrees d^e and d^o, runs the two opposite-parity QSVT sequences and sums them by an LCU: a mixed-parity circuit. Each sequence uses the unitary dilation

$$U(\tilde{H}) = \begin{pmatrix} \tilde{H} & \sqrt{I - \tilde{H}\tilde{H}^\dagger} \\ \sqrt{I - \tilde{H}^\dagger\tilde{H}} & -\tilde{H}^\dagger \end{pmatrix}, \tag{K.1}$$

Table 20: The paper's solutions at its printed resolution: L^∞ error of the paper-faithful and corrected variants, and the corrected variant's best over n. Classical computation.

Fig.	Problem	Printed n	Faithful	Corrected	Corrected, best over n
3b	R2a	3	1.2×10^{-3}	1.2×10^{-3}	3.2×10^{-13} $(n = 5)$
3c	R2b	3	$1.8 \times 10^{-2\dagger}$	$1.8 \times 10^{-2\dagger}$	2.3×10^{-11} $(n = 5)$
3d	R2c	5	2.2×10^{-3}	2.2×10^{-3}	5.6×10^{-12} $(n = 6)$
4a	R3a	3	1.9×10^{-15}	1.9×10^{-15}	1.9×10^{-15} $(n = 3)$
4b	R3b	3	3.0×10^{-15}	3.0×10^{-15}	2.3×10^{-15} $(n = 4)$
4c	R3c	5	$8.7 \times 10^{-1\dagger}$	4.0×10^{-14}	4.9×10^{-15} $(n = 3)$
4d	R3d	5	$1.2^\dagger$	1.5×10^{-13}	8.0×10^{-15} $(n = 3)$
5a	R4a	3	3.1×10^{-3}	1.8×10^{-6}	3.8×10^{-14} $(n = 4)$
5b	R4b	3	1.9×10^{-2}	7.8×10^{-3}	8.5×10^{-12} $(n = 5)$
5c	R4c	4	7.2×10^{-3}	4.9×10^{-10}	1.6×10^{-13} $(n = 5)$
5d	R4d	4	6.1×10^{-2}	4.9×10^{-7}	5.1×10^{-12} $(n = 5)$
6b	R5a	3	3.9×10^{-3}	1.4×10^{-4}	2.1×10^{-13} $(n = 4)$
6d	R5b	4	1.9×10^{-4}	4.7×10^{-5}	4.6×10^{-12} $(n = 5)$
6f	R5c	5	8.5×10^{-8}	7.3×10^{-8}	6.1×10^{-11} $(n = 6)$
7b	R6a	2	$2.2 \times 10^{-16\dagger}$	$2.2 \times 10^{-16\dagger}$	–
7c	R6b	2	$6.8 \times 10^{-3\dagger}$	$6.8 \times 10^{-3\dagger}$	–

Table 21: Which printed choice causes the Fig. 5 error: the paper-faithful variant over $n = 4, \ldots, 10$, each correction alone, and both together.

Fig.	Problem	Faithful	Exact root	Chebyshev source	Both, best
5a	R4a	3.1×10^{-3}	–	–	3.8×10^{-14} $(n = 4)$
5b	R4b	1.6×10^{-2}	4.0×10^{-2}	3.6×10^{-2}	8.5×10^{-12} $(n = 5)$
5c	R4c	7.2×10^{-3}	4.8×10^{-3}	1.0×10^{-2}	1.6×10^{-13} $(n = 5)$
5d	R4d	6.1×10^{-2}	8.1×10^{-2}	1.1×10^{-1}	5.1×10^{-12} $(n = 5)$

Table 22: The claims of Wu et al. tested in this thesis, and their verdicts.

	Claim	Verdict
A-01	$\mathbb{G}$, $\mathbb{M}$, $\mathbb{n}_1$ and $\mathbb{B}$ as printed at $n = 2$	reproduced
A-02	the printed η_e of Figs 3–5 (11 panels)	within one unit of the last printed digit; 9/11 equal when rounded
A-03	the printed η_e of Fig. 6	not reproduced
A-04	the printed η_e of Fig. 7	not reproduced
A-05	the printed Laplace solution	a typographical error (cosh for cos)
A-06	the $\mathbb{B}$ circuit (Fig. 2): $n + 2$ qubits, prefactor $\sqrt{2^{n+1}}$	confirmed
A-07	the $\mathbb{G}$ circuit (Fig. 9): $2n + 3$ qubits, $O(n^3)$ gates, error decreasing exponentially	–
A-08	the $\mathbb{D}^{(0)}$ circuit (Fig. 8): prefactor $\sqrt{2^{n+1}}/f(x_s)$	confirmed
A-09	the linear H is positive definite with distinct eigenvalues	–
A-10	the doubled-space H has at least 2^{2n-1} zero eigenvalues	–
A-11	the time rule $t \geq \lambda_{\max}/(\lambda_2(n-1))$ suffices	–
A-12	the necessary terms scale as $O(\mathrm{poly}(n))$ and are efficient	true for gates; not for the subnormalisation
A-13	the examples are solved accurately at the printed n	distorted where rounded points or Maclaurin sources enter

built from PREPARE and SELECT subroutines over the unitary decomposition of $\tilde{H}$ with $m = \lceil \log_2 L \rceil$ ancilla for L unitaries. The time satisfies the approximate bound $t \geq \lambda_{\max}/(\lambda_2(n-1))$ (Equation (3.7)). For the paper's two Legendre examples of its supplement, $t = 15$ and $t = 8$ recover the ground state with $(d^e, d^o) = (6, 7)$. Beyond the $O(1/\Delta)\,\mathrm{polylog}(1/\varepsilon)$ preparation time, the paper expects algorithmic thermalisation to reach $O(1/\sqrt{\Delta})$, and notes that GQSP, GQET and GQSVT could improve the circuit.

K.6 The tau-truncation ablation

Table 23: Dropping the top k rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to the paper's least squares.

Problem	n	Least-squares L^∞	σ_0	Tau-truncated L^∞	σ_0
R2a	5	3.2×10^{-13}	3.0×10^{-11}	1.6×10^{-13}	0
R2a	6	2.2×10^{-12}	4.9×10^{-10}	3.8×10^{-12}	0
R2a	7	2.8×10^{-11}	7.8×10^{-9}	1.0×10^{-10}	0
R2b	5	2.3×10^{-11}	5.3×10^{-11}	3.7×10^{-12}	0
R2b	6	1.8×10^{-10}	8.5×10^{-10}	7.4×10^{-11}	0
R2b	7	9.8×10^{-10}	1.4×10^{-8}	8.3×10^{-10}	0
R2c	5	2.2×10^{-3}	1.6×10^{-2}	1.8×10^{-3}	0
R2c	6	5.6×10^{-12}	5.5×10^{-10}	1.8×10^{-11}	0
R2c	7	2.4×10^{-10}	1.1×10^{-8}	1.9×10^{-10}	0

L Software and Reproducibility

L.1 Architecture

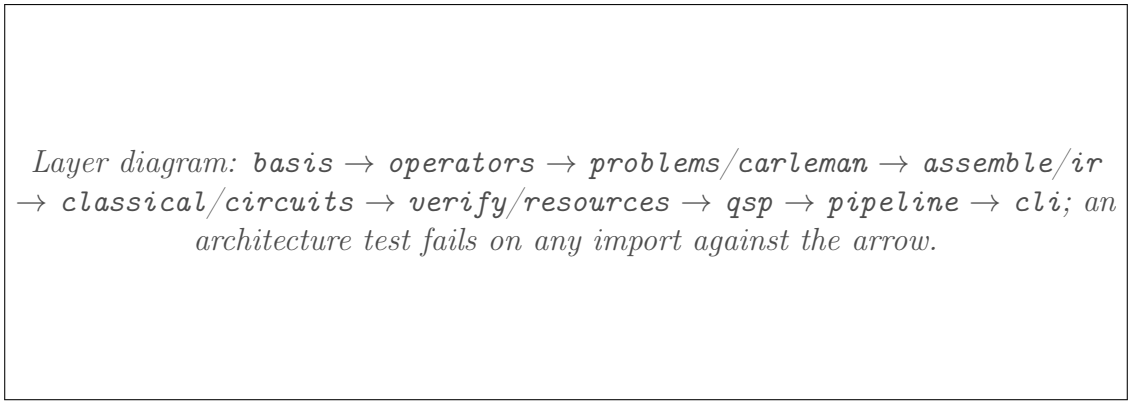


Figure 15: The layers of the `pihm` package and the one direction in which they may import.

L.2 Records and provenance

L.3 Campaigns and computational resources

Table 24: Resource classes for the Setonix campaigns.

Class	Request	Typical runs
S	`work`, 1 CPU, 4 GB, 30 min	construction, IR checks, small exact verification
M	`work`, 16 CPUs, 64 GB, 4 h	exact verification to 20 qubits; mid-size exact emulation
L	`work`, 64–128 CPUs, ∼230 GB, 12–24 h	circuit-simulation ceilings; large emulation; probes to 30 qubits
XL	`highmem`, 1 TB	the largest emulation and probe runs

L.4 Tests

Table 25: The test tiers of the `pihm` package.

Tier	Content	Run
Unit and property	operator identities over random n and x; primitives; arithmetic; IR; filters; phases	every commit
Golden	fixtures from the frozen legacy code and the paper's printed $n = 2$ matrices	every commit
Reproduction	the paper's targets, both variants	every commit
Integration	every problem, both forms, smallest n, all stages	every commit
Documentation	Sphinx build, doctests, spelling	every commit
Claims	one test per claim; writes the claims report	at gates
Slow	larger-n verification; circuit-simulation ceilings	at gates, on Setonix

L.5 Regenerating the results

L.6 Environment

M Research Proposal

Table 26: Deviations from the research proposal of May 2025.

Proposed	Outcome	Reason
Extending QSVT to the generalised construction for complex-valued, non-Hermitian operators		
Nonlinear PDEs: Burgers, nonlinear Schrödinger, complex Ginzburg–Landau		
Data-informed Hamiltonians (not proposed)		

Masters Research Proposal

Dhyan Patel [23340229]
The University of Western Australia

May 25, 2025

Title: Solving Nonlinear PDEs via Quantum Singular Value Transformation

Supervisor: Professor Jingbo Wang

1 Introduction

1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

2 Background

2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

2.2 Literature Review

2.2.1 Exploration of Problems

Burgers Equation (Viscous)

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u\frac{\partial u}{\partial x}$.

Nonlinear Schrödinger Equation

$$i\frac{\partial \psi}{\partial t} + \nabla^2\psi + |\psi|^2\psi = 0 \tag{2}$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2\psi$.

Complex Ginzburg-Landau Equation

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha)\nabla^2 A - (1 + i\beta)|A|^2 A \tag{3}$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix A.

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values Σ of a matrix A block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \tag{4}$$

Assuming we have access to such a unitary U and the ability to apply a sequence of controlled-phase rotations by an angle ϕ, we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[\prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd Poly}^{(\text{SV})}(A) \\ \left[\prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even Poly}^{(\text{SV})}(A) \end{cases} \tag{5}$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \ldots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree d [1]. $\tilde{\Pi}_\phi$ and Π_ϕ represent a controlled phase rotation by ϕ where $\tilde{\Pi}$ and Π are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of A.

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

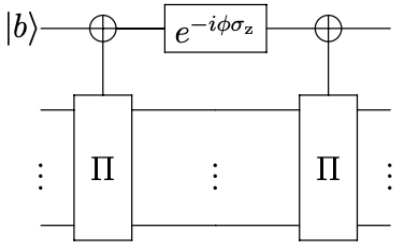


Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, Π_ϕ [2]

2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \qquad x \in [a, b]. \tag{6}$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s, namely

$$u(x) = s\langle T(x)|\psi\rangle \tag{7}$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + af(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^{\intercal 2} + a\mathbb{I}. \tag{8}$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx}\langle T(x)|\psi\rangle = \langle T(x)|D^\intercal|\psi\rangle. \tag{9}$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on x. For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\intercal$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\intercal \mathbb{A} + \mathbb{B}^\intercal \mathbb{B}. \tag{10}$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \tag{11}$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. H is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace's equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\intercal 2} \otimes \mathbb{G}_y^{\intercal 2}. \tag{12}$$

For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we're able to solve.

Example Solutions:

Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t,x)}{\partial t} - k\frac{\partial^2 f(t,x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0,x) = \sin(2\pi x), f(t,\pm 1) = 0\} \tag{13}$$



Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4\frac{d^2 f(x)}{dx^2} + 2\left(\frac{df(x)}{dx}\right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \tag{14}$$

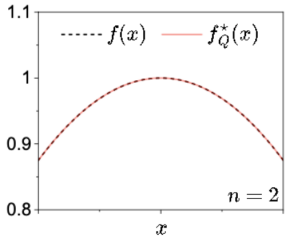


Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{fx^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \tag{15}$$

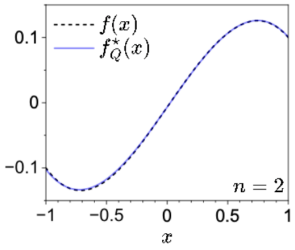


Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as n on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

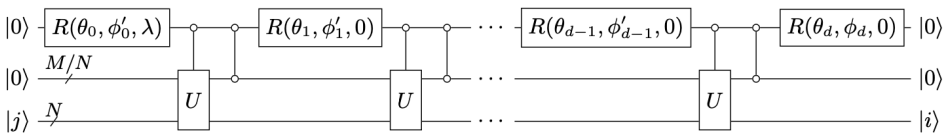


Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix A, block encoded as U. R is a rotation matrix analogous to Π for QSVT [9].

3 Research Plan

3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *Pennylane* and results will be compared to classical solving algorithms. Efforts wil be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

Proposed Plan:

1. Reproduce the Current Physics-Informed QSVT Method:
   (a) Implement the standard QSVT framework in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
   (a) Create the GQSVT algorithm in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

Timeline:

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work in underway on implementing the current physics-informed Hamiltonian approach using QSVT.

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